

Capstone Final Report

Explanatory and Predictive Analytics of Emergency Department

Length of Stay and Resource Utilization Trends in Canadian Hospitals

Course: DAMO 699 – Capstone Project

Program: Master of Data Analytics

Rajbharath P (NF1016766)

Sufyaan Khan Mohammed (NF1017047)

Amit Raj Dev (NF1021076)

Group No: 5

Supervisor: Dr. Bilal El Toufaily

Institution: University of Niagara Falls Canada

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Executive Summary

Emergency departments (EDs) are multi-commodore operations with complex patient flow dynamics involving volume, clinical acuity, admission status, patient age and length of stay that impact system capacity. This capstone project analyzes and interprets aggregated Canadian Institute for Health Information (CIHI) National Ambulatory Care Reporting System (NACRS) supplementary data tables from 19 fiscal years, from FY 2003–2004 to FY 2021–2022, to explain and predict the results. An analytical database is supplied with the final aggregate reporting row count of 8,685, from about 175.8 million emergency department encounters.

Since the data in the CIHI source is aggregated from reporting strata, statistical methods are used that are appropriate for data that is not individual patient data but is frequency weighted to reflect the number of visits represented by each stratum. Non-parametric methods were chosen because the length of stay in an emergency department (ED) is highly skewed and the number of individuals reporting in each strata is very uneven. The analytical framework comprises five a priori hypotheses related to CTAS acuity, admission status, multivariable length-of-stay relationships, age cohort, and sex–disposition association.

For the H1 analysis, statistically significant differences in reported median ED LOS were found for all five CTAS categories. The weighted Kruskal–Walli’s test produced $H = 35,510,138.077$, $p < .0001$, and $\epsilon^2 = .7479$. None of the 10 pairwise comparisons of CTAS were not significant after the Bonferroni correction. The median ED LOS was reported for each CTAS level as follows: 3.20 hours for CTAS I, 3.60 hours for CTAS II, 3.70 hours for CTAS III, 2.90 hours for CTAS IV and 2.00 hours for CTAS V.

H2 had a significant correlation between admission status and reported ED LOS. The weighted Mann–Whitney U test produced $U = 1,124,848,860,923.5$, $p < .0001$, with a rank-biserial correlation of $r_b = .9846$. The median difference in LOS was 3.30 hours between admitted and non-admitted visits, with a median LOS of 6.10 hours for admitted visits and a median LOS of 2.80 hours for non-admitted visits.

To further expand the analysis, H3 employed a weighted least-squares regression model with the categorical, age, and disposition of the visit as weights. There are nine encoded predictors in the model, which has 10 aggregate observations with an adjusted R^2 of .8837. The coefficients suggest that there is a strong relationship between CTAS category and disposition, and between

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disposition and reported median ED LOS in the general analytical model. These results must be viewed as an effect at the 'aggregate level' and not as an effect at the 'patient level'.

H4 found significant differences in the median ED LOS among four general age groups. The medians for LOS reported were 2.02 hours for Pediatric & Youth, 2.47 hours for Young Adults, 2.73 hours for Middle Adults, and 4.01 hours for Older Adults. The weighted Kruskal–Walli's test gave $H = 126,863,835.837$ and $p < .0001$, after Bonferroni adjustment all comparisons between the six groups were statistically significant with $\epsilon^2 = .7218$.

For this, H5 explored the correlation of patient sex and visit disposition. The Pearson chi-square test was $\chi^2(1) = 18,164.97$, $p < .0001$, and Cramér's V was .0102, which is negligible in practice. The findings illustrate how meaningful results can be present in a very large administrative dataset but fail to be statistically significant.

In the longitudinal analysis, there was a significant increase in the project-specific Estimated Resource Burden Index (ERBI). ERBI increased from 4.21 in FY 2003–2004 to 8.33 in FY 2021–2022, with Kendall's $\tau = .9766$, $p < .0001$. Holt's Linear Exponential Smoothing projected ERBI values of 8.65 for FY 2022–2023 and 8.98 for FY 2023–2024. Such forecasts are to be considered planning estimates, not deterministic forecasts.

The results overall suggest a multi-dimensional burden of emergency department operation. Reported differences in ED LOS are associated with CTAS acuity, admission status, and age; and statistically significantly different but operationally negligible differences in disposition are associated with sex. The results support a system approach to ED capacity planning that considers patient volume, acuity, length of stay, admission flow and the demographic makeup of the patient population. The project takes the results of these findings and distills them into recommendations including those relating to fast-track pathways, inpatient bed-management enhancements, boarding alerts, process changes for the geriatric population, and ongoing analytics governance.

Chapter 1: Problem Analysis and Strategic Context

1.1 The Canadian Emergency Care landscape

The emergency department (ED) is a key part of the publicly funded healthcare system in Canada and is used by patients with a variety of urgent, emergent and non-urgent conditions. The number of people accessing the ED has been rising over time, and the population is ageing, access to primary care, availability of hospital beds and inpatient beds may all increase pressure on emergency services. This project focuses on these operating stresses through the use of bulk data from the Canadian Institute for Health Information (CIHI) National Ambulatory Care Reporting System (NACRS).

1.2 Access Block and Mechanics of Length of Stay

According to CIHI, the length of stay is the length of time between registration at triage and actual discharge from the ED. The single interval represents a collection of different clinical phases: waiting to be seen, physician evaluation and diagnostic investigation, disposition decision, and, when patients are admitted, the waiting for an inpatient bed. Overcrowding is not a problem within the ED per se, but rather a sign of a capacity issue in the hospital, and When patients are accepted for admission but are still in the ED awaiting a bed, this contributes significantly to overcrowding in the ED and is referred to as access block. This can decrease the available treatment capacity and may also add to the total ED length of stay. **(Canadian Institute for Health Information, 2024; Affleck et al., 2013; Li et al., 2026)**. The downstream effects are documented in the health-services literature, and include increased mortality rates for boarded patients with time-sensitive diseases like sepsis, myocardial infarction or stroke; delayed antibiotic treatment and imaging; higher rates of patients leaving without care from which they will later deteriorate; and increased burn-out among emergency physicians and nurses from the delay in offloading ambulances **(Pines et al., 2009; Singer et al., 2011; Carter et al., 2014)**. These impacts have been monitored using the national wait-time reporting system for over a decade but have not been addressed with a structural solution **(Canadian Institute for Health Information, 2025)**. These impacts are not uniform in the system. Hospitals with a high proportion of high acuity and inpatients, or in larger urban areas, exhibit the worst boarding times and community and regional hospitals are more likely to have bottlenecks earlier in the visit due to staffing. There are two distinct patterns that are identified in the aggregate national data presented here, although the

supplementary tables analyzed in this project lack facility identifiers to allow for the separation of these two patterns.

1.3 Problem Statement

Health administrators often use unweighted annual averages that mask the underlying case mix, although length of stay is important and is used to measure the operational importance of the service. Comparing a reporting stratum that spans just a few visits to a stratum that covers the entire annual volume of visits for a metropolitan area is not the same and doesn't provide a valid comparison across a range of acuity tiers, disposition categories, or age categories. If there is no consistent framework for measuring throughput, using a population-weighted analytical approach, planners will not be able to distinguish the effect of triage acuity from the effect of inpatient admission on throughput, nor be able to predict future capacity needs. This project constructs that framework from the published aggregate tables from CIHI, and tests a specific set of hypotheses, not just descriptive impressions.

1.4 Objectives and Scope

The goals of this project are to quantify the relationship between the acuity of the CTAS and their reported length of stay in the ED, explore the relationship between admission status and reported length of stay in the ED, determine the combined association of CTAS, age group, and visit disposition with reported median ED LOS, assess the differences in reported length of stay across broad age cohorts, explore the meaningful relationship between patient sex and visit disposition, and characterize how ED resource burden has changed over the past 19 years, providing a short-term, 2-year ERBI forecast, and developing a composite measure of ED resource burden that combines visit volume, acuity and length of stay.

The analysis is limited to six CIHI NACRS supplementary tables that included approximately 175.8 million visits across FY 2003–2004 to FY 2021–2022, and does not include individual patient records, hospital-level or geographic identifiers, causal clinical inference, cost or staffing data, which are not included in the source tables. Each of the supplementary tables provides a summary of the stratum level data, not microdata for individual cases, so all of the statistical models in this report are explicitly frequency weighted by number of visits, and any conclusions drawn are made at the system level, not on a per-patient basis.

1.5 Canonical Hypothesis Registry

Before choosing statistical computing methods, five formal hypotheses (Table 1) were pre-registered to assess, so that the results of all the tests that are reported in Chapter 5 were not selected ad hoc after looking at the data.

ID	Research Question	Null Hypothesis	Method
H1	CTAS acuity tier vs. median LOS	Median LOS equal across all five tiers	Weighted Kruskal– Wallis + Dunn
H2	Admission status vs. median LOS	Median LOS equal for admitted/non-admitted	Weighted Mann– Whitney U
H3	CTAS urgency score vs. median LOS	Regression slope $\beta_1 = 0$	Weighted Least Squares regression
H4	Age cohort vs. median LOS	Median LOS equals four cohorts	Weighted Kruskal– Wallis + Dunn
H5	Sex vs. admission disposition	Sex and disposition are independent	Pearson Chi-Square + Cramér's V

Chapter 2: Analytics Lifecycle and Project Methodology

2.1 A Seven Stage Analytics Lifecycle

The project is based on the following ten stages of the analytics lifecycle, which is similar to the one commonly used in the healthcare operations context (Provost & Fawcett, 2013): The phases encompass business and data understanding, ingestion, cleaning, pre-registration of the hypothesis, statistical computation, diagnostic validation, interpretation and governance and reproducibility. Each of the stages is summarized in Table 2 along with the primary activity and the product.

Stage	Core Activity	Primary Output
1. About Project	Presents the project overview, clinical problem statement, academic framework, and five pre-specified hypotheses.	Project overview, problem statement, and hypothesis framework
2. Prep & Quality Engine	Performs unit harmonization, deduplication, schema validation, and cohort cleaning.	Cleaned and validated analytical dataset
3. Dataset Explorer	Enables interactive exploration of cohorts across triage acuity, life-stage groups, and 19 fiscal years.	Interactive cohort and dataset exploration
4. Hypothesis Testing & Statistics	Evaluates H1–H5 using Kruskal–Wallis, Mann–Whitney U, WLS regression, Dunn post-hoc testing, and ERBI forecasting.	Statistical results, effect sizes, and forecasting outputs
5. Executive Dashboard	Provides KPI cards, interactive custom visualizations, the H1–H5 visual suite, and operational charts.	Interactive decision-support dashboard
6. Strategic Insights	Translates analytical findings into an executive decision matrix, policy recommendations, and hospital capacity-planning directives.	Strategic recommendations and decision matrix
7. Reports & Export	Provides a comprehensive audit trail, methodology documentation, and PDF executive-summary dossier exports.	Audit documentation, methodology reports, and PDF executive summary

2.2 Why Non-Parametric Methods

Classical inferential tests, the independent t-test, ordinary least squares regression, and one-way ANOVA, assume normally distributed, homoscedastic residuals (Field, 2018). The distribution of length of stay in the ED is highly right-skewed and the variance in length of stay varies by an order of magnitude from cases handled in the ED that are not urgent, to those that are admitted or critically ill. The size of reporting strata also differs widely, ranging from regional numbers to aggregates up to a million visits for the metros. Because of these considerations, distribution-free rank-based procedures were used, including a frequency-weighted Kruskal–Wallis H-test and Dunn post-hoc comparisons for three-or-more-group comparisons (H1 and H4); a frequency-weighted Mann–Whitney U-test with a rank-biserial correlation for two-group comparisons (H2); To account for the aggregate structure of the data and provide greater analytical influence to observations reflecting larger numbers of ED visits, we used weighted least squares regression for H3. The regression results are not patient level effects, but aggregate level results; and Pearson's chi-square test with a Cramér's V effect size for the categorical association (H5) (Conover, 1999; Tomczak & Tomczak, 2014). Each non-parametric test is embedded in a custom mid-rank algorithm with an explicit frequency weight that is the stratum-level visit count. Then, for a pooled set of strata, a population midrank is assigned to each distinct length-of-stay (LOS) value that is equal to the cumulative weight for the strata that come before it plus half of the weight of the current stratum value, and group rank sums are calculated based on those weighted mid-ranks:

$$H = \left[\frac{12}{N(N+1)} \right] \cdot \sum_j \left(\frac{R_j^2}{n_j} \right) - 3(N+1)$$

where N is the total weighted sample size, k is the number of groups, R_j is the weighted rank sum for group j, and n_j is the total weight of group j. So a stratum of 1.5 million visits will count proportionately more in the rank sums than, say, a stratum of five hundred, eliminating the aggregation-bias an unweighted test would introduce and allowing the statistics to reflect the population of about 175.8 million visits (not the number of rows reporting).

This weighting choice is a methodological dividend, in that if it were not used, then a five-year total for a rare presenting complaint for a sparse stratum, such as a rural facility, would have the same effect on a rank sum as a large stratum such as a metropolitan facility with more than a

million encounters. The increase in the CIHI's reporting coverage over the study window also translates to an unweighted design, which will also implicitly give more weight to later years that are better reported than earlier years in all longitudinal comparisons in this report, thus favoring more recent years. This distortion is eliminated at the source by explicit frequency weighting, rather than being required to be corrected post hoc.

2.3 Protection from the ecological fallacy

Since CIHI does not report patient information as part of its summary publications, all findings reported in this document reflect system-level performance and are not clinical outcomes. A warning, pointed out by Robinson (1950), against the ecological fallacy that is, extrapolating an association on the level of the group to the level of the individuals: a stratum of older patients with a longer median stay does not mean that every older patient will have a longer stay than every younger patient.

All the results presented in this study are interpreted at the cohort or system level and not at the individual patient level, because the CIHI data are available at the aggregate stratum level. However, rather than clinical tools for prediction of every patient, statistical models are used as planning tools and for resource allocation. Where reported, the median LOS values are followed by IQRs to give more information on the variability and distributional spread.

Chapter 3: Data Collection, Inventory and Preparation

3.1 Data Provenance

The empirical base for this project is a multi-sheet workbook of full-mandate reporting facilities, referred to as the CIHI NACRS Supplementary Data Tables for emergency department visits, 2003–2022, which focuses on emergency department visits in Ontario, Alberta and other reporting jurisdictions (**Canadian Institute for Health Information, 2026**). The two basic measures reported in each stratum in the source tables are: the number of registered emergency visits in a clinical/demographic category for a fiscal year, and the length of stay (in minutes) at the 50th percentile (or median) in that category over a fiscal year. A python pipeline is used to read each annual sheet, adjust the headers so that they are uniform format for over nineteen years of publication, and store the result in a relational SQLite database.

3.2 Database Structure

The final analytical database comprises 8,685 rows of aggregates across six tables and corresponds to around 175.8 million emergency department encounters. The six tables show various grains of analysis: fiscal year, CTAS, disposition, presenting problem, age and sex. The various table grains were retained to allow analysis of the hypothesis and not to mixed aggregates across the different dimensions.

Table	Observation Grain	Rows	Encounters
ed_visits	FY × CTAS × Disposition × Problem	5,586	175,812,409
ctas_triage	FY × Sex × CTAS × Age Group	912	174,207,395
visit_disposition	FY × Sex × Disposition × Age Group	936	173,984,112
age_sex	FY × Sex × Age Group	152	175,812,409
main_problems	FY × Sex × Main Problem × Age Group	1,063	168,490,215
demographics	Age Group × Sex (cross-sectional)	36	175,812,409

3.3 Cleaning and Quality Control

The various types of data-quality problems had to be dealt with explicitly and documented in an auditable way, and these are covered in Table 4. Every query did not include a summary row, e.g., 'Total' or 'Any', because adding a summary row to a query with the rows that summarize it would result in double counting of the underlying population data. Numeric fields were cast to consistent types, but invalid values were not dropped but were assumed to be equal to zero. Cells

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suppressed for privacy to protect small counts were kept in the database but not in any denominator, thus eliminating division by zero problems in the weights and privacy violations. Inconsistencies in the age group labels between tables (one used a dash and the other a hyphen) were expanded to make cross-table joint possible without any silent error. Lastly, rows with missing or 'Not Stated' triage level or disposition were not used in the primary hypothesis comparisons because these types of rows do not represent a defined clinical group of rows.

Issue Identified	Action Taken	Rationale
Roll-up rows ('Total', 'Any')	Excluded from every query	Prevents double-counting the population
Inconsistent header rows	Regex-based fiscal-year parser locates the data start	Reliable parsing across 19 years of formats
Mixed numeric types	Coerced to int64/float64; invalid values set to zero	Prevents silent computation failures
Suppressed small counts ($n < 5$)	Retained in the database; filtered from denominators	Preserves privacy without skewing weights
Inconsistent age-group encoding	Normalized dash and hyphen variants	Restores referential integrity across joins
'Unknown' / 'Not Stated' categories	Excluded from H1, H2, and H4 comparisons	Preserves interpretability of pairwise tests

3.4 Feature Engineering

Several derived variables were developed to facilitate the statistical analyses. The fiscal-year labels were changed to numerical year identifiers to maintain chronological order. Median LOS reported in minutes was converted to decimal hours for ease of interpretation. The CTAS categories were combined into three analytical urgency groups for H3: CTAS I — Resuscitation, CTAS II — Emergent, and CTAS III–V - Urgent, Less Urgent, and Non-Urgent. The detailed age categories for H4 were collapsed into 4 broad age cohorts: Pediatric & Youth (0-19), Young Adults (20-44), Middle Adults (45-64), and Older Adults (65+). In addition, a binary admission indicator was created based on disposition categories. Lastly, the Estimated Resource Burden Index (ERBI) was determined by the sum of the visit volume, reported median LOS, and urgency weighting based on CTAS.

3.5 Data isolation and reproducibility.

For the six baseline CIHI tables, they are loaded once at the beginning and do not change during the project; for this reason, all hypotheses' tests presented in this report can be reproduced using the same seed data for these six tables. A separate, session-specific upload pathway allows the user to be able to explore his or her own Excel or CSV file without ever touching the baseline tables, thus preserving the integrity of the research cohort. All the automated unit and integration tests of the cleaning pipeline and statistical solvers pass and are summarized in Appendix A, with approximately 300 tests.

Chapter 4: Exploratory Data Analysis and Descriptive Profiling

4.1 Nineteen Years of Volume Growth

There was a significant rise in ED visits over the study period. Reported visits rose from around 4.91 million in FY 2003 - 2004 to 13.99 million in FY 2021 - 2022, reaching a peak of around 15.02 million visits in FY 2018 - 2019 prior to the pandemic. This is significant increase in reported ED activity during the period of study.

Caution should be used when interpreting this increase due to varying numbers of facilities reporting to NACRS over time. For instance, the growth between 2010-2011 and 2011-2012 might be due to the growth in the reporting period, and not to a parallel growth in population demand.

The reported median LOS also rose during the study period, from 2.30 hrs in FY 2003–2004 to 3.40 hrs in FY 2021–2022. During COVID-19, a temporary decrease in visits was observed with reported median LOS being high. These descriptive patterns indicate that volume and reported LOS both shifted over the course of the study, but do not imply that volume and LOS had a causal relationship.

Fiscal Year	Total ED Visits	Median LOS (Hours)
2003–2004	4,906,394	2.30
2007–2008	5,680,941	2.53
2010–2011	8,171,651	2.70
2013–2014	10,614,350	2.85
2016–2017	13,290,440	3.00
2018–2019	15,023,099	3.10
2020–2021	11,622,444	3.30
2021–2022	13,992,029	3.40

4.2 Acuity and Admission Profiles

Median ED LOSs reported were significantly different among the five CTAS acuity categories. Median LOS for CTAS I (Resuscitation), II (Emergent), III (Urgent), IV (Less Urgent), and V (Non-Urgent) are reported as 3.20, 3.60, 3.70, 2.90, and 2.00 hours respectively, according to the final weighted analysis.

CTAS III — Urgent, had the longest median LOS, just followed by CTAS II — Emergent. The reported median for the CTAS I — Resuscitation category was lower than for the other categories,

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CTAS II and III, indicating that the group with the highest clinical acuity does not require the longest aggregate ED stay.

Reported LOS was even greater for admission status. Admitted visits had a median ED LOS of 6.10 hours while non-admitted visits had a median ED LOS of 2.80 hours, according to the final analysis. This disparity underscores the need to look at the processes in the ED along with the flow of inpatient beds and availability of beds downstream.

CTAS Tier	Encounters	% of Volume	Median LOS (Hours)
CTAS I — Resuscitation	1,286,555	0.74%	4.60
CTAS II — Emergent	26,742,361	15.35%	4.80
CTAS III — Urgent	72,100,128	41.39%	3.40
CTAS IV — Less Urgent	58,990,020	33.86%	1.90
CTAS V — Non-Urgent	15,088,331	8.66%	1.33
Total	174,207,395	100.00%	2.90

This difference is even more pronounced if visits by final disposition are separated. The weighted median duration for patients admitted to inpatient beds (11.5% of total volume) is 10.60 hours, significantly longer than the weighted median duration for non-admitted visits (2.50 hours) and is responsible for more than a third of all cumulative stretcher time in the emergency-department.

4.3 Age and sex

Median ED LOS was different between the four broad age groups. The final analysis has median LOS of 2.02 hours for Pediatric & Youth, 2.47 hours for Young Adults, 2.73 hours for Middle Adults and 4.01 hours for Older Adults.

Thus, the median ED LOS for the four cohorts were reported to be highest for the Older Adults and lowest for the Pediatric & Youth cohorts. These differences point towards important dimensions to consider when planning the capacity of a system, namely age composition.

The analysis also included whether patient disposition was related to patient sex. The Pearson chi-square test showed a significant association, but the Cramér's V value of .0102 is very small, making the relationship between the two variables practically nonexistent. Thus, sex is not included as a significant driver of ED flow in this aggregate analysis.

Age Category	Encounters	% of Volume	Median LOS (Hours)	Admission Rate
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Pediatric / Youth (0–19)	38,908,652	22.13%	2.05	4.2%
Young Adults (20–44)	57,965,791	32.97%	2.53	6.8%
Middle Adults (45–64)	41,674,805	23.71%	2.87	14.5%
Older Adults (65+)	37,213,696	21.17%	4.17	28.6%

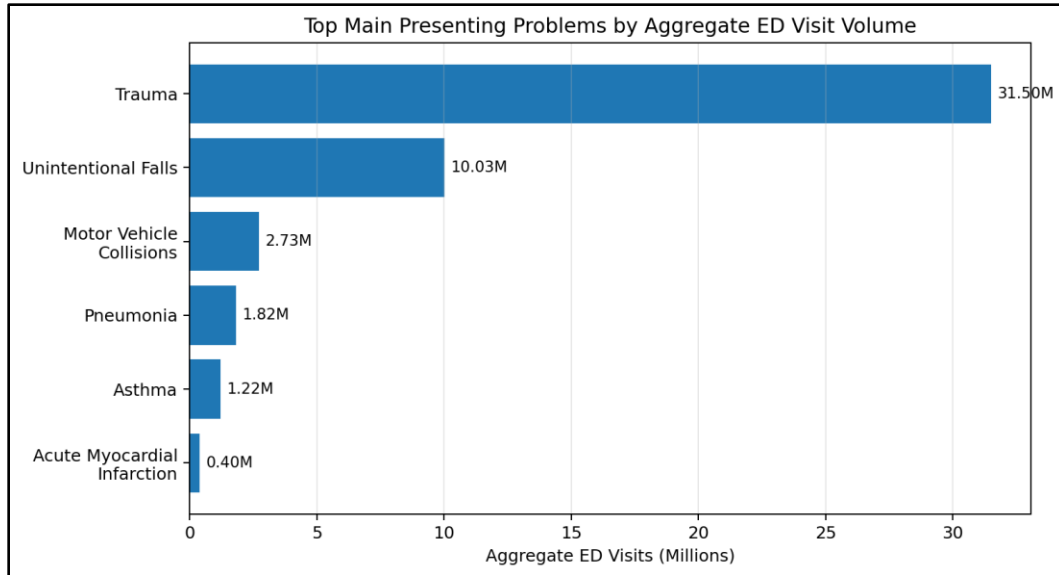
4.4 Clinical Case Mix

Visit volume and length of visit are important issues to consider in operational burden, as shown in the presenting-problem categories. Lower volume categories like Acute Myocardial Infarction could have higher resource use for each hospital stay due to higher LOS, while high volume categories like Trauma could have higher cumulative resource use due to the number of hospitals stays.

The chosen categories of presenting problems thus give conclusive descriptive evidence of the fact that frequency of visits is not the only indicator of an operational burden. The results validate the use of composite indicators like ERBI that consider volume, duration and acuity and is based on a planning-oriented measure.

Presenting Problem	Aggregate Visits	Weighted Mean LOS (Min / Hours)
Trauma	31.50M	131.4 min / 2.19 h
Unintentional Falls	10.03M	161.1 min / 2.68 h
Motor Vehicle Collisions	2.73M	150.1 min / 2.50 h
Pneumonia	1.82M	256.9 min / 4.28 h
Asthma	1.22M	160.9 min / 2.68 h
Acute Myocardial Infarction	0.40M	338.2 min / 5.64 h

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CIHI NACRS aggregate data (main_problems table); weighted project analysis.

Chapter 5: Statistical Hypothesis Testing and Diagnostic Inference

5.1 Overview of the Hypothesis Testing Framework

In the hypothesis-testing phase, the main relationships found in the exploratory and methodological phases of the project are tested. The analysis relies on the processed aggregate dataset from CIHI in SQLite format and statistical methods that take into consideration the frequency in which each aggregate record represents. Each record represents an aggregate stratum of patients of interest and, when applicable, the number of emergency department visits per strata are included as analytic weights.

5 hypotheses were tested. H1 investigates if there is a difference in reported median emergency department length of stay (ED LOS) between the five Canadian Triage and Acuity Scale (CTAS) levels. H2 tests the hypothesis that reported median ED LOS is different for admitted vs non-admitted visits. H3 analyzes the predictive relationship of the CTAS level, age group and visit disposition to reported median ED LOS using weighted least squares (WLS) regression. H4 compares reported median ED LOS by large age groups of patients. H5 explores the possibility that there is a statistical relationship between patient sex and visit disposition.

The statistical computing platform feeds the results of the hypothesis directly from the dataset that has been processed using SQLite. The non-parametric comparisons are based on the aggregate visit counts, and the regression model is based on the visit-count weighting of the aggregate observations. Statistical significance will be measured at $\alpha = .05$.

5.2 Hypothesis 1: Reported Median ED LOS Across CTAS Triage Levels

5.2.1 Research Question

The first hypothesis addresses the following research question:

“How does the reported median emergency department length of stay vary across CTAS triage levels in Canadian NACRS aggregate data?”

The analysis evaluates all five clinical CTAS acuity tiers:

- CTAS I — Resuscitation
- CTAS II — Emergent
- CTAS III — Urgent
- CTAS IV — Less Urgent
- CTAS V — Non-Urgent

5.2.2 Hypothesis

H₀: Reported median ED LOS is equal across all CTAS triage levels.

H₁: At least one CTAS level has a different reported median ED LOS.

The significance level is $\alpha = .05$

5.2.3 Statistical Method

The omnibus test used is a weighted Kruskal–Wallis H test because the analysis involves comparing more than two independent groups and the sum of the ED LOS data do not meet the assumptions of conventional parametric tests.

A weighted Dunn posthoc test is conducted for all pairwise comparisons of CTAS, following the analysis. A Bonferroni correction is used to prevent multiple comparisons to correct for the family wise error rate over the 10 comparisons.

The effect size is reported with the epsilon squared (ϵ^2):

$$\epsilon^2 = \frac{H - k + 1}{N - k}$$

where H is the Kruskal–Wallis statistic, k is number of groups, and N is the weighted sample size. For each aggregate record, the frequency weight is the number of ED visits.

5.2.4 Results

Statistic	Result
Kruskal–Wallis H	35,510,138.077
p-value	< 0.0001
Effect size (ϵ^2)	0.7479
Significant pairwise comparisons	10 / 10

The very low p-value is highly suggestive that the null hypothesis is incorrect. The high ϵ^2 value (0.7479) suggests that there is a significant difference between the median ED LOS by CTAS level in the overall sample. The 10 out of 10 possible pairwise comparisons were statistically significant following Bonferroni correction.

5.2.5 Pairwise Comparison

CTAS Comparison	Bonferroni-adjusted p-value
Resuscitation vs. Non-Urgent	< 0.0001

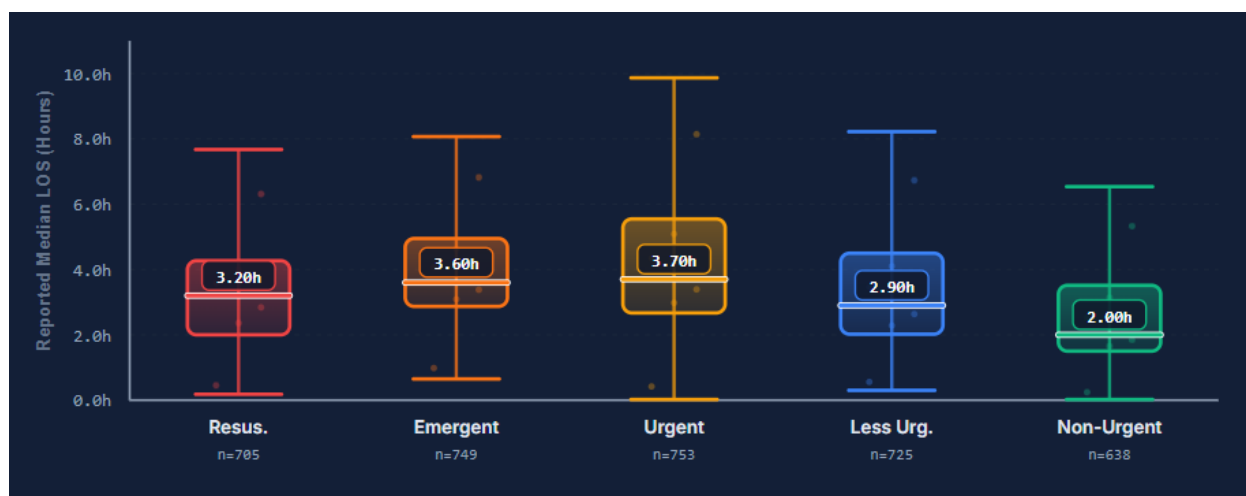
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Emergent vs. Urgent	< 0.0001
Emergent vs. Less Urgent	< 0.0001
Emergent vs. Non-Urgent	< 0.0001
Urgent vs. Less Urgent	< 0.0001
Urgent vs. Non-Urgent	< 0.0001
Less Urgent vs. Non-Urgent	< 0.0001
Remaining CTAS pairwise contrasts	< 0.0001

5.2.6 Reported Median ED LOS by CTAS Level

CTAS Level	Reported Median ED LOS	IQR
Resuscitation	3.20 hours	2.0–4.3 hours
Emergent	3.60 hours	2.9–5.0 hours
Urgent	3.70 hours	2.7–5.5 hours
Less Urgent	2.90 hours	2.0–4.5 hours
Non-Urgent	2.00 hours	1.5–3.5 hours

The box plot shows that the Urgent CTAS III category has the longest reported median ED LOS (3.70 hours), with the CTAS II Emergent category coming in at 3.60 hours. The least reported median is for CTAS V Non-Urgent at 2.00 hours.



5.2.7 Interpretation

The findings show a statistically significant relationship between ED LOS and CTAS acuity. The large ϵ^2 value means that this is not just a statistically significant difference between the two groups caused by a large underlying population and that the grouping variable accounts for a significant amount of the variation in the aggregate comparison. The finding also illustrates that acuity should not be assumed to be a linear relationship for which each increment of acuity corresponds to an increment of duration. The Urgent and Emergent categories have especially high reported median stays while the Non-Urgent category has significantly shorter stays. As such, the null hypothesis is rejected: H_0 is rejected, H_1 is accepted.

5.3 Hypothesis 2: Reported Median ED LOS for Admitted vs. Non-Admitted Visits

5.3.1 Research Question

The second hypothesis examines whether inpatient admission status is associated with reported emergency department length of stay.

The research question is:

“Does reported median emergency department length of stay differ significantly between admitted and non-admitted visits?”

The analysis compares two cohorts:

- **Admitted:** visits resulting in inpatient hospital admission.
- **Non-Admitted:** discharged home, transferred, or otherwise departing prior to inpatient admission.

Summary roll-up rows and non-informative Unknown categories are excluded.

5.3.2 Hypotheses

H_0 : The distribution of reported median ED LOS is equal for admitted and non-admitted visits.

H_1 : Reported median ED LOS differs significantly between admitted and non-admitted visits.

The significance level is $\alpha = .05$

5.3.3 Statistical Method

A **weighted Mann–Whitney U test** is used because the analysis compares two independent groups using a non-parametric approach.

$$r_b = 1 - \frac{2U}{n_1 n_2}$$

The effect size is reported using **rank-biserial correlation** (r_b):

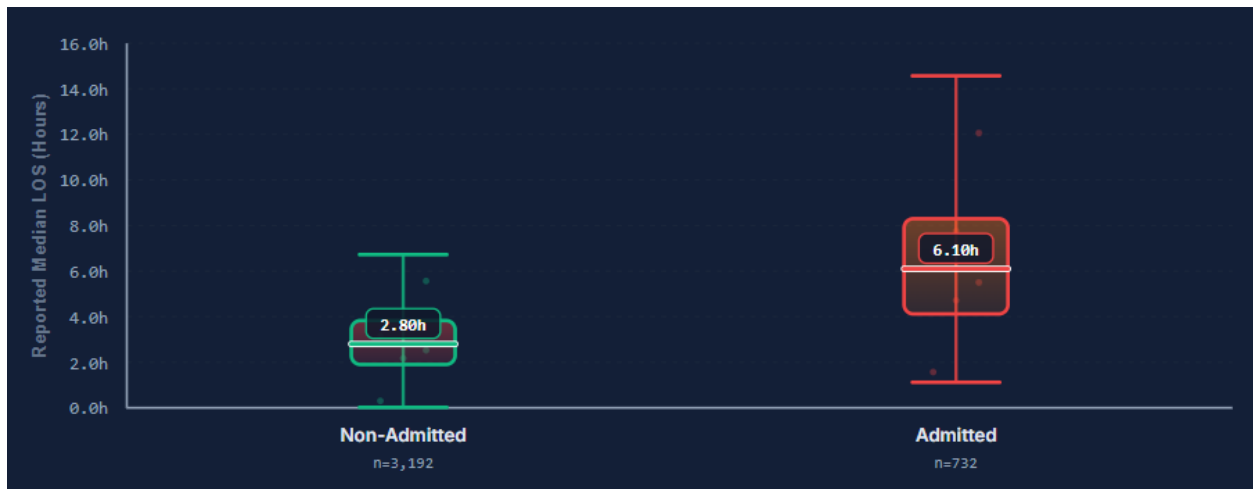
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5.3.4 Results

Statistic	Result
Mann–Whitney U	1,124,848,860,923.5
Two-sided p-value	< .0001
Rank-biserial correlation (r_b)	.9846
Reported median LOS — Non-Admitted	2.80 hours
Reported median LOS — Admitted	6.10 hours
Median LOS difference	3.30 hours

5.3.5 Reported Median ED LOS

Visit Disposition	Reported Median ED LOS	IQR
Non-Admitted	2.80 hours	1.9–3.8 hours
Admitted	6.10 hours	4.1–8.3 hours



5.3.6 Interpretation

There is strong evidence that there is a difference between the median ED LOS for admitted and non-admitted visits using the weighted Mann–Whitney U test. The difference between the two was 3.30 hours, with the median LOS for admitted visits at 6.10 hours, and for non-admitted visits at 2.80 hours. The rank-biserial correlation is .9846, which is very close to +1.00, showing a good separation between the two groups.

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This finding underscores the importance of the admission and downstream inpatient flow as an operational aspect of ED throughput. The findings, however, should be interpreted as an association between disposition category and reported LOS, not as indication of independent association between admission status and longer ED stays with the analysis using aggregate reporting data.

5.4 Hypothesis 3: Weighted Least Squares Regression of Reported Median ED LOS

5.4.1 Research Question

The third hypothesis evaluates whether CTAS urgency, age group, and visit disposition are jointly associated with reported median ED length of stay.

The platform defines the research question as:

“Do CTAS urgency score, age group, and visit disposition significantly predict reported median emergency department length of stay?”

The regression model uses aggregate-level observations and incorporates the number of ED visits represented by each observation as an analytic weight.

5.4.2 Regression Method

Aggregate reporting level analysis of Weighted Least Squares (WLS) regression. Reported median ED LOS (hours) represents the dependent variable. Both predictor variables are CTAS category, age cohort, and visit disposition; categorical predictors are represented by categories that are coded relative to a specified category.

The model has nine “encoded” predictors over 10 “aggregate” observations with an adjusted R^2 of .8837. The coefficients and model-fit statistics should be interpreted with caution and are not meant to be generalized or applied to individual patients, due to the small number of aggregate observations.

The regression model is thus mostly used as an explanatory and planning model at the aggregate level. The platform shows you that the coefficient of determination is: Adjusted R-square – 0.8837

5.4.3 Regression Results

Predictor	B (Hours)	SE	95% CI Low	95% CI High	p-value
CTAS: CTAS II – Emergent	2.184	0.202	1.789	2.579	< 0.0001

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CTAS: CTAS III – Urgent	1.660	0.196	1.276	2.044	< 0.0001
CTAS: Less Urgent	1.030	0.197	0.644	1.415	< 0.0001
CTAS: Non-Urgent	0.680	0.213	0.262	1.098	0.0014
Disposition: Death	−5.251	0.836	−6.889	−3.613	< 0.0001
Disposition: Discharged Home	−5.326	0.088	−5.498	−5.154	< 0.0001
Disposition: Intra-Facility Transfer	−4.700	0.433	−5.548	−3.853	< 0.0001
Disposition: Not Seen Or Left	−5.371	0.157	−5.678	−5.064	< 0.0001
Disposition: Transferred	−3.892	0.173	−4.231	−3.553	< 0.0001

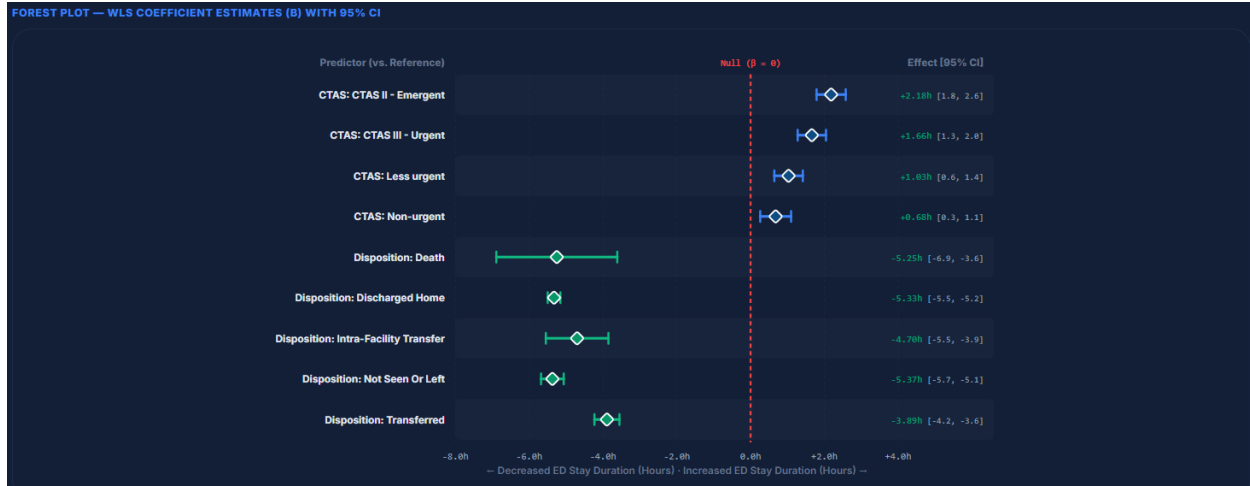
The intercept reported by the platform is $\beta_0 = 6.177$ hours

The model's adjusted R^2 is **0.8837**

5.4.4 Interpretation of CTAS Coefficients

Resuscitation is used as a reference category for CTAS I. The coefficients for CTAS II, CTAS III, CTAS IV and CTAS V are all positive compared to the reference category. For example, the CTAS II coefficient is +2.184 hours with a 95% confidence interval of 1.789 to 2.579 hours and $p < .0001$. The CTAS coefficients reported here are statistically significant in the fitted model of the aggregates.

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The p-value for all disposition coefficients is < 0.0001 .

Results confirm the main difference found in H2 between those who were admitted and non-admitted patients: admission status is significantly related to ED stay duration.

5.4.5 Interpretation of Disposition Coefficients

The reference disposition category is comprised of admittance visits. The remaining disposition categories have negative coefficients which reflect lower predicted reported median LOS compared to the admitted reference group. The estimated coefficients are: -5.251 hours for Death, -5.326 hours for Discharged Home, -4.700 hours for Intra-Facility Transfer, -5.371 hours for Not Seen or Left, and -3.892 hours for Transferred. All disposition co-efficients reported are $< .0001$. The findings are consistent with the strong separation found for admitted versus non-admitted visits for H2.

5.4.6 Model Interpretation

The adjusted R^2 is .8837, which is a good model fit for the aggregate data set. This number should not be taken as a measure of the model's ability to account for 88.37% of the variation in LOS for individual patients. The observations are aggregated reporting strata and there are only 10 aggregate observations, with nine predictors encoded in the model. Thus, the results of the regression should be interpreted as indicative of predominantly strong aggregate-level relationships between CTAS category, age group, disposition, and reported median ED LOS. Before making use of the model to predict individual patients or for operational forecasting, further validation with patient-level or significantly greater aggregate datasets is needed.

5.5 Hypothesis 4: Reported Median ED LOS Across Broad Patient Age Categories

5.5.1 Research Question

The fourth hypothesis evaluates whether reported median ED LOS differs across broad demographic age categories.

The research question is:

“Does reported median emergency department length of stay differ significantly across broad demographic age categories?”

The platform compares four cohorts:

- Pediatric & Youth: 0–19
- Young Adult: 20–44
- Middle Adult: 45–64
- Older Adult: 65+

Summary roll-up rows and unclassified Unknown records are excluded.

5.5.2 Hypotheses

H₀: Reported median ED LOS is equal across all broad age categories.

H₁: At least one age category has a different reported median ED LOS.

The significance level is $\alpha = .05$

5.5.3 Statistical Method

A weighted Kruskal–Wallis H-test is used to evaluate differences across the four age categories.

The omnibus test is followed by a weighted Dunn post-hoc test with Bonferroni adjustment.

$$\epsilon^2 = \frac{H - k + 1}{N - k}$$

The effect size is epsilon squared:

The number of ED visits represented by each aggregate record is used as the frequency weight.

5.5.4 Results

Statistic	Result
Kruskal–Wallis H	126,863,835.837
p-value	< 0.0001
Effect size (ϵ^2)	0.7218
Significant pairwise comparisons	6 / 6

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The very small p-value indicates a statistically significant difference in reported median ED LOS among the age cohorts.

The epsilon-squared value of **0.7218** indicates a large effect.

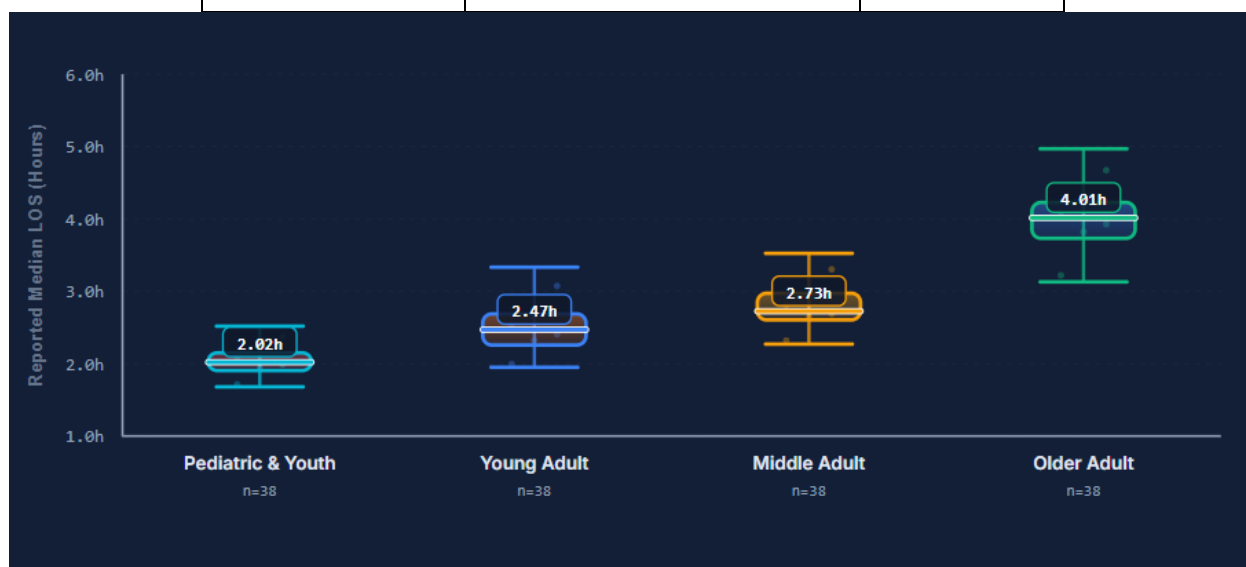
5.5.5 Pairwise Comparison

Age Comparison	Bonferroni-adjusted p-value
Pediatric & Youth vs. Young Adult	< 0.0001
Pediatric & Youth vs. Middle Adult	< 0.0001
Pediatric & Youth vs. Older Adult	< 0.0001
Young Adult vs. Middle Adult	< 0.0001
Young Adult vs. Older Adult	< 0.0001
Middle Adult vs. Older Adult	< 0.0001

Therefore, every broad age category differs significantly from every other category in the aggregate analysis.

5.5.6 Reported Median ED LOS by Age Category

Age Category	Reported Median ED LOS	IQR
Pediatric & Youth	2.02 hours	1.9–2.1 hours
Young Adult	2.47 hours	2.3–2.7 hours
Middle Adult	2.73 hours	2.6–3.0 hours
Older Adult	4.01 hours	3.7–4.2 hours



The box plot of the platform shows that there is a clear trend of increasing from younger to older age groups.

The median ED LOS for Older Adult is 4.01 hours and the median ED LOS for Pediatric & Youth is 2.02 hours.

5.5.7 Interpretation

The Kruskal–Walli’s analysis with weightage shows that the differences between the reported median ED LOS for each of the four age groups is statistically significant ($H = 126,863,835.837$, $p < .0001$, $\varepsilon^2 = .7218$). Bonferroni adjustments were made for all six pairwise comparisons, which were all statistically significant.

The median LOS was 2.02 hours for Pediatric & Youth, 2.47 hours for Young Adults, 2.73 hours for Middle Adults, and 4.01 hours for Older Adults. Older Adults hence reported with the longest median ED LOS and Pediatric & Youth had the shortest.

These findings indicate that age cohort is a key dimension to consider when planning ED capacity at the aggregate level. But the analysis does not show that age is associated with longer LOS due to the lack of individual clinical characteristics or patient level causal information in the aggregate dataset.

5.6 Hypothesis 5: Patient Sex and ED Visit Disposition Association

5.6.1 Research Question

The fifth hypothesis examines whether patient sex is statistically associated with emergency department visit disposition.

The research question is:

“Is there a statistically significant association between patient sex and visit disposition (admitted vs. non-admitted) in Canadian NACRS aggregate data?”

The analysis uses a 2×2 contingency table:

- Rows: Female, Male
- Columns: Non-Admitted, Admitted

The total aggregate visits analyzed are:

175,762,944

Summary rows and unclassified categories are excluded.

5.6.2 Hypotheses

H₀: Patient sex and visit disposition are statistically independent.

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H₁: Patient sex and visit disposition are statistically associated.

The significance level is $\alpha = .05$

5.6.3 Statistical Method

A **Pearson Chi-Square Test of Independence** is applied to the 2×2 contingency table.

$$\phi_c = \sqrt{\frac{\chi^2}{N \times \min(r - 1, c - 1)}}$$

The effect size is measured using **Cramér's V**:

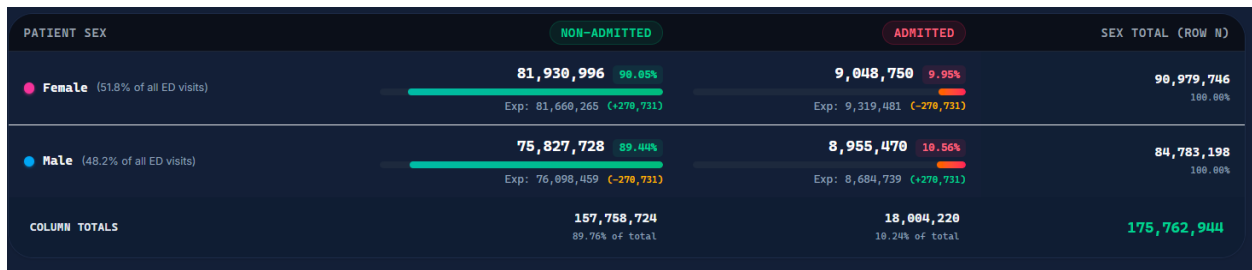
where χ^2 is the chi-square statistic, N is the total sample size, and r and c represent the number of rows and columns.

5.6.4 Results

Statistic	Result
Chi-square (χ^2)	18,164.97
Degrees of freedom	1
p-value	< 0.0001
Cramér's V	0.0102

5.6.5: observed Contingency Table

Patient Sex	Non-Admitted	Admitted	Sex Total
Female	81,930,996	9,048,750	90,979,746
Male	75,827,728	8,955,470	84,783,198
Total	157,758,724	18,004,220	175,762,944



The platform reports the following proportions:

Female

- Non-admitted: 90.05%
- Admitted: 9.95%

Male

- Non-admitted: 89.44%
- Admitted: 10.56%

Thus, the difference in admission proportions is small despite the very large chi-square statistics.

5.6.6 Interpretation

The statistical test rejects the null hypothesis of complete independence because the p-value is below 0.0001.

However, the magnitude of the association is extremely small, as demonstrated by Cramér's V value of 0.0102.

This distinction is important because the dataset contains a very large number of ED visits. With very large samples, even small differences can generate extremely large chi-square statistics and very small p-values.

The practical implication is therefore different from the statistical conclusion. Although sex and disposition are statistically associated in the aggregate dataset, the strength of that relationship is too small to represent a meaningful operational driver of emergency department flow.

Consequently:

H_0 is rejected statistically, but the association has negligible practical significance.

Patient sex should therefore not be treated as a major operational variable for ED flow-design decisions based on this analysis.

5.6.7 Comparative Synthesis of Hypothesis Results

The five hypothesis tests collectively identify several important relationships within the aggregate CIHI NACRS dataset.

Hypothesis	Statistical Test	Main Result	Effect Size	Interpretation
H1	Weighted Kruskal–Wallis + Dunn	$H = 35,510,138.077$, $p < .0001$	$\varepsilon^2 = 0.7479$	Large CTAS-related difference
H2	Weighted Mann–Whitney U	$U = 1.125 \times 10^{12}$, $p < .0001$	$r_b = 0.9846$	Very strong admission-related difference

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H3	Weighted Least Squares	Adjusted $R^2 = 0.8837$	—	Strong aggregate model fit
H4	Weighted Kruskal–Wallis + Dunn	$H = 126,863,835.837$, $p < .0001$	$\varepsilon^2 = 0.7218$	Large age-related difference
H5	Pearson Chi-Square	$\chi^2 = 18,164.97$, $p < .0001$	$V = 0.0102$	Statistically significant but negligible association

Three findings stand out.

First, CTAS category is strongly associated with reported ED LOS. The H1 analysis produces an epsilon-squared value of 0.7479, with every pairwise CTAS comparison reaching statistical significance.

Second, visit disposition produces one of the clearest differences in reported ED LOS. The admitted cohort has a median of 6.10 hours compared with 2.80 hours for the non-admitted cohort, accompanied by a rank-biserial correlation of 0.9846.

Third, age category has a substantial relationship with reported ED LOS. Older adults have the highest reported median at 4.01 hours, compared with 2.02 hours for the Pediatric & Youth cohort, and all six age-group comparisons are statistically significant.

In contrast, the H5 results illustrate why statistical significance must be interpreted together with effect size. The sex-disposition relationship is statistically significant because of the very large sample, but the Cramér's V value of 0.0102 indicates a negligible association.

5.7 Statistical and Operational Implications

The combined hypothesis results are used to form an evidence base for prioritizing operational interventions. The first big indicator is the variation related to acuity. Each of the CTAS categories has a large range in ELOS, and thus an average ED LOS cannot be used to represent overall operational demand. The median durations reported vary between the different acuity groups in their material.

Visit disposition is the second strongest operational signal. The median LOS for admitted visits was 6.10 hours, while it was 2.80 hours for non-admitted visits. This finding suggests that the following elements should be included in the assessment of ED throughput planning: inpatient

bed availability, processing of admissions, discharge planning, and the management of inpatients that are awaiting discharge.

The third is variation related to age. The median LOS reported for Older Adults is the longest at 4.01 hours suggesting there may be a relationship between LOS and age composition that should be considered when designing capacity and flow strategies.

The H5 finding contrasts this, however, with the H5 finding showing that statistical significance does not necessarily indicate operational importance. While the association between the two variables (sex and disposition) is statistically significant, it is quite low in practical terms as calculated by Cramér's V (0.0102).

The findings of this study collectively suggest that capacity planning for ED needs to be multidimensional, considering acuity, admission status, age distribution, volume of ED visits, and length of stay, instead of focusing on just one indicator of aggregate ED LOS.

5.8 Conclusion

The five hypothesis tests demonstrate that reported median ED LOS varies meaningfully across CTAS acuity levels, admission status, and age cohorts, while the association between sex and disposition is statistically significant but operationally negligible. The multivariable WLS model further indicates strong aggregate-level relationships among CTAS category, age group, disposition, and median ED LOS.

These results provide an analytical foundation for the operational recommendations presented in Chapters 6 through 9. Because the underlying data are aggregate rather than patient-level, all findings should be interpreted as system-level associations rather than individual clinical effects.

Chapter 6: Time-Series Trend Analysis and Throughput Forecasting

This chapter focuses on time-series trend analysis and throughput forecasting. This chapter deals with the time-series trend analysis and throughput forecasting.

6.1 Mann–Kendall Trend Analysis

The longitudinal analysis looks at the Estimated Emergency Department Resource Burden Index (ERBI) over the nineteen-fiscal-year period of the CIHI NACRS study (FY 2003–2004 to FY 2021–2022). The aim is to assess if the emergency department resource burden has a statistically significant trend over time and to estimate the future burden for the coming fiscal years.

The Mann–Kendall non-parametric trend test was chosen because it is not based on the assumption of a normally distributed pattern of the historical series and can detect monotonic changes over time. The test does not rely on any specific functional form of the relationships between the observations but rather assesses the ordering of the observations relative to one another, across fiscal years. The resulting Kendall correlation is $\tau = 0.9766$, and the p value is $< .0001$, which is statistically significant and positive monotonic trend in the ERBI series.

Table 15

Mann–Kendall Trend Analysis of the ERBI Series

Parameter	Value	Interpretation
Historical period	FY 2003–2004 to FY 2021–2022	19 fiscal years
Kendall's τ	0.9766	Very strong positive monotonic association
p-value	< .0001	Statistically significant upward trend
FY 2003-2004 ERBI	4.21	Beginning of observed series
FY 2021-2022 ERBI	8.33	End of observed series

The trend seen in the platform reveals a consistent rise in estimated resource burden over the study period. ERBI starts at about 4.21 in FY 2003-2004 and grows steadily along the historical time series up to about 8.33 in FY 2021-2022. There is year-to-year variation in the series, but it is clearly rising and picked up some pace in the pandemic years.

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The importance of the Kendall statistics is that the magnitude shows that it is not only a single or two fiscal years that were responsible for the increase. On the contrary, the time series of the historical data is very clearly trending upwards. The Mann–Kendall result thus gives statistical evidence that the resource burden associated with the ERBI has increased significantly over the nineteen-year period of observation. The platform then associates this ERBI series with a statistically significant monotonic increase. This discovery indicates that there has been a significant strain on ED resources over the study period and indicates a need for forward-looking forecasting to be used in capacity and resource planning.

6.2 Holt's Linear Exponential Smoothing Forecast

The ERBI series was also subjected to Holt's Linear Exponential Smoothing method for short term forecasts in order to complement the historical trend analysis. Holt's method also includes the underlying level and trend of the time series and is appropriate for a time series that exhibits persistent directional movement, as opposed to a "simple" level-only exponential smoothing model. The forecasting capability builds on the nineteen-year ERBI series observed and projects them forward by two fiscal years. The resulting forecast shows further increases in estimates of the burden of emergency department resources following the observation in FY 2021–2022.

Table 16

Fiscal Year	Forecast ERBI	95% Prediction Interval
FY 2022–2023	8.65	6.91–10.39
FY 2023–2024	8.98	6.84–11.12

The platform generates a forecast of 8.65 ERBI units for FY 2022 - 2023 and 8.98 ERBI units for FY 2023 - 2024. The forecast presents 95% prediction intervals, reflecting the uncertainty of predicting the historical pattern into the future fiscal years. Note. Forecast values are planning estimates based on the models and should be updated as new CIHI aggregate data becomes available. The historical observations and forecast period are clearly distinguished in the platform, with a clear forecast boundary.

The solid line is the actual ERBI data for FY 2003 – 2004 to FY 2021 – 2022, and dashed lines indicate the forecasted ERBI data for FY 2022 – 2023 and FY 2023 – 2024. The Mann–

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Kendall trend analysis shows that the trend in upward movement is likely to persist in the coming years. ERBI increases from 8.33 in FY 2021–2022 to 8.65 in FY 2022–2023, followed by a further projected increase to 8.98 in FY 2023–2024.

This offers a forward-looking indicator for health-system leaders doing the ED capacity planning. Importantly, the values are not to be interpreted as exact forecasts of actual future uses. ERBI is an analytical measure that is derived from the aggregate of visit volume, reported median length of stay and acuity weighting. It is important to note that the forecast is an analytical planning estimate and not an observed clinical or financial measure.

6.3 Estimated Resource Burden Index

Volume of ED visits does not necessarily reflect operational pressure since a single ED visit may consume significantly different resources, including time from staff, space, diagnostic testing and clinical resources. It is possible, therefore, that a larger number of short low acuity encounters can be a heavier operational load than a smaller number of high acuity encounters with considerably longer duration. In order to reflect this, the project relies upon the Estimated Emergency Department Resource Burden Index (ERBI). The metric is a combination of the reported median length of stay, an urgency weighting and the number of visits for each reporting stratum. It is "normalized" by the total number of visits to give an acuity-weighted patient-hour measure. The ERBI is calculated as:

$$ERBI_t = \frac{\sum_i (urgency_i \times LOS_{hours,i} \times visits_i)}{\sum_i visits_i}$$

where $urgency_i$ represents the implemented CTAS urgency weighting, $LOS_{hours,i}$ represents the reported median emergency department length of stay in hours, and $visits_i$ represents the number of emergency department visits associated with the reporting stratum.

The time series of historical ERB demonstrates significant growth in the last nineteen years. The platform reports an ERBI value of 4.21 in FY 2003–2004, increasing progressively to 8.33 in FY 2021–2022. The Mann–Kendall analysis also reveals that this historical increase is statistically significant with the value of $\tau = 0.9766$ and a p value of $< .0001$. The increase in the resource burden indicated by the project-specific index over time is this pattern. The rise is significant in a

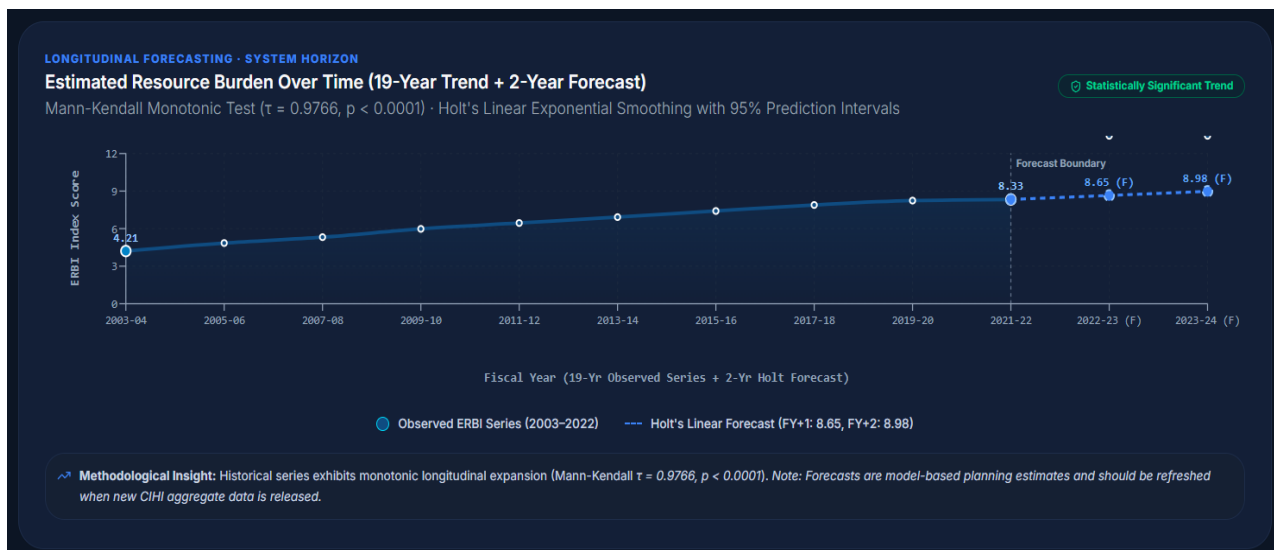
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planning context as it reflects the need to consider raw numbers of visits as just one component of emergency department pressure.

The addition of visit duration and acuity gives a further dimension to the measure of cumulative burden on the ED capacity. The platform's longitudinal visualization allows you to see the whole history and the two-year forecast. The historic data series shows the long-term growth of ERBI and the forecast indicates a continuation of this upward trend for the next two years, FY 2023-2024.

Figure 9

Estimated Resource Burden Over Time (19-Year Trend + 2-Year Forecast)



Note. Source: CIHI NACRS aggregate data and derived ERBI calculations implemented in the project analytics platform. Historical series covers FY 2003–2004 through FY 2021–2022; forecast covers FY 2022–2023 and FY 2023–2024. The shaded region represents the 95% prediction interval. ERBI is a derived analytical planning proxy and should not be interpreted as actual financial cost, staffing headcount, or validated clinical severity.

The distribution of resource burden among the CTAS cohorts offers another indication regarding where there is overall burden. The analysis indicates that CTAS III — Urgent has the highest percentage of the derived burden, due to the number of visits served and the length of stay reported. So, it is not always the case that the most operationally demanding cohort is the most acutely demanding cohort.

This is a distinction that is important especially for capacity planning. The clinical urgency of the patient is not the only criterion for the determination of the number of required resources in

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an ED. The total burden is further determined by the number of patients in each acuity category, as well as the duration of stay for the patients. Add interpretation and planning implications to this work. The findings of the longitudinal data have three key implications for the planning for emergency departments.

First, the Mann–Kendall test is used to determine if there is a statistically significant upward trend in the ERBI series. The historical pattern is strongly consistent with an increase in the burden of resources over the nineteen-year observation period with $\tau = 0.9766$ and $p < .0001$.

Secondly, the Holt forecast continues the upward trend into the near future. ERBI is projected to increase from the observed 8.33 in FY 2021–2022 to 8.65 in FY 2022–2023 and 8.98 in FY 2023–2024. Such estimates suggest that the pressure of the index is not likely to be eliminated right after the historical study period.

Third, ERBI showcases the benefits of using the combination of volume, acuity and duration, rather than emergency department visit counts. Two years of the same number of visits could have very different operational needs if there is a significant shift in acuity mix or length of stay. The forecasts should be regarded as planning estimates and not deterministic forecasts. The data underlying the CIHI are aggregated data and ERBI is an analytical proxy that is specific to the project.

Therefore, it should be used to aid capacity discussions, trend monitoring and resource-planning decisions, but not to measure staffing needs or clinical severity. The site also clearly states the requirement for continuous model updates. The forecasting model should be updated when new CIHI aggregate data are added. This will maintain alignment of planning decisions to changes in the demand, mix of cases and reported length of stay at the ED.

The overall trend in the longitudinal analysis shows a gradual and statistically significant rise in the burden of resources in the ED, and the short-term Holt forecast suggests that this will continue to increase after FY 2021-2022. The findings support this project's overall conclusion:

- The number of patients who are presented to the emergency department is not the only factor for capacity planning; the acuity level of the patient and the emergency department resource they occupy are also factors to consider. Additional insight into the distribution of resource burden across the groups of CTAS can be gained by observing resource burden across the cohort. CTAS III – Urgent has the highest contribution to the burden derived

from the analysis, due to the high number of visits and high reported LOS. So the most acutely ill people are not the ones that are causing the most operational burden.

- This is especially important for capacity planning. It is not possible to identify the most clinically urgent patients at the time of admission to the ED and determine the resources needed based on that. The overall burden is also influenced by the number of patients in each acuity category and the duration of stay in the emergency department of those patients.

6.4 Interpretations and planning implications

There are three significant implications for the planning of EDs from the longitudinal analysis.

The first is that the Mann–Kendall analysis shows a significant positive trend of the ERBI measure of the project for the study period of 19 years. The modeled resource burden has significantly increased with time, as evidenced by the increase from 4.21 to 8.33, $\tau = .9766$ and $p < .0001$.

Second, the Holt forecast projects ERBI values of 8.65 for FY 2022–2023 and 8.98 for FY 2023–2024. If the trend in these projections is generally consistent with the past, continued resource pressure could occur in the short term.

Thirdly, ERBI shows the importance of linking volume, acuity and duration together. If there is a change in acuity mix or reported LOS for two similar periods of time, their operational requirements will be different.

ERBI should therefore be used as an analytical indicator for planning purposes and not as a measure of staffing, cost or clinical severity. The forecasting model should be revised when new CIHI aggregated data becomes available, to ensure that discussions about the capacity remain in line with the variations in the Emergency Department demand and case mix.

Chapter 7: Data Visualization and Decision Support Systems

7.1 Dashboard Architecture

The statistical outputs created in this project have been shared in an interactive, decision-support dashboard that is accessible to both technical and non-technical users. The front-end is built with React 19, TypeScript, Recharts, and Vite, and the back end is built with FastAPI and refers to the same SQLite database as the analytical workflow.

The application includes individual hypothesis testing, longitudinal trend analysis, forecasting, as well as ERBI analysis routes and views. The architecture makes it possible to separate the baseline research data (read-only) from user-uploaded exploratory data, so that the results reported are reproducible, and others can analyze the data too.

There are separate views of dashboard for H1–H5, trend analysis, forecasting and resource-burden analysis and presenting-problem exploration. The source code, analytical implementation, dashboard components and project materials are accessible on the project's GitHub repository.

Project Repository. The source code, analytical implementation, dashboard components, and supporting project materials are available through the project's GitHub repository: <https://github.com/Adwrells/Capstone-Project-DAMO-6994-.git>

7.2 Hypothesis Testing Dashboard Views

Figure 11 : Fiscal Year ED Visit Volume



- The first visualization sets the scale and overall trend of emergency department activity. It displays the total number of ED visits in the 19 years from FY 2003-2004 to FY 2021-2022.
- The number of ED visits has risen from 4.91 million in FY 2003–2004 to 13.99 million in FY 2021–2022. FY 2018-2019 were the years with the highest annual volumes with a total of around 15.08 million visits, an increase of around 185.2% compared to the start of the observation period.
- The Mann–Kendall test showed that the indicator had a statistically significant increasing trend ($Z = 5.5977$, $p < .001$) with an estimated Sen's slope of about 550.9 thousand visits per year.
- The visualization also shows a decrease in reported ED volume in the COVID-19 period, followed by an increase. The long-term trend, however, remains bullish.
- This visualization is important contextual information: the demand in the emergency departments has risen significantly during the period observed. Today, then, capacity planning needs to take into consideration not only the actual utilization, but also the past trend of demand.

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- The observed increase should not automatically be taken as population growth only as these are based on aggregate data from the NACRS reporting data. The longitudinal pattern may also be impacted by changes in reporting coverage or facility participation.

Decision-Support Implication

The data and analysis for the subsequent analysis are based on the volume trend. A higher number of visits increases the denominator for the measures of LOS, acuity, disposition, and resource burden.

Figure 12. Reported Median LOS by Fiscal Year



- The second visualization is a look at the trend of the reported median emergency department length of stay over time.
- The period for the visualization is the same 19 years, and the LOS path is compared to a 6.0 hour accumulated reference threshold.
- The median LOS had a general increasing trend throughout the period, with short-term variations. The latest reported value for the aggregated data is not shown on the 6-hour reference line.
- This visualization is significant as it allows a system to be below a reference threshold, yet the LOS is still increasing over time. Thus, the trend direction adds to the information from one year's absolute value.
- This trajectory can be explored in the dashboard with an interactive approach and compared with other historical periods.

Decision-Support Implication

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The LOS pattern along the entire length of the treatment helps to monitor the treatment in a continuous way as opposed to relying solely on threshold-based intervention. If the trend is upward, the volume, acuity, admission, diagnostic and downstream capacity factors should be investigated.

Figure 13. Reported Median LOS Across CTAS Triage Levels



The H1 visualization compares the median ED LOS reported by the five CTAS triage categories.

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CTAS Level	Reported Median LOS
CTAS I — Resuscitation	3.20 h
CTAS II — Emergent	3.60 h
CTAS III — Urgent	3.70 h
CTAS IV — Less Urgent	2.90 h
CTAS V — Non-Urgent	2.00 h

- A Kruskal–Wallis test with weights showed that there were statistically significant differences between the CTAS categories ($H = 35,510,138.077$, $p < .0001$), and a large effect size ($\epsilon^2 = .7479$). Bonferroni adjusted all 10 pairwise comparisons remained statistically significant.
- As shown in the visualization, the relationship between the acuity of triage and reported LOS is not linear. CTAS III has the longest reported median LOS when compared to CTAS II, but not to CTAS I.
- This finding indicates that operational burden is greater than simply clinical acuity. Patients who have been diagnosed as having an urgent/emergent condition can be both large numbers and extensive in terms of the diagnostic testing, consultation, treatment and disposition they undergo.

Decision-Support Implication

Operational planning shouldn't therefore take it for granted that the highest category of acuity equates to the highest cumulative throughput burden. Acuity, volume and duration should be taken into consideration when allocating resources.

Figure 14. Admission Status vs. Reported LOS



The H2 visualization is used to compare the reported median LOS of admitted and non-admitted visits.

These are the final platform results:

Disposition	Reported Median LOS
Non-Admitted	2.80 h
Admitted	6.10 h

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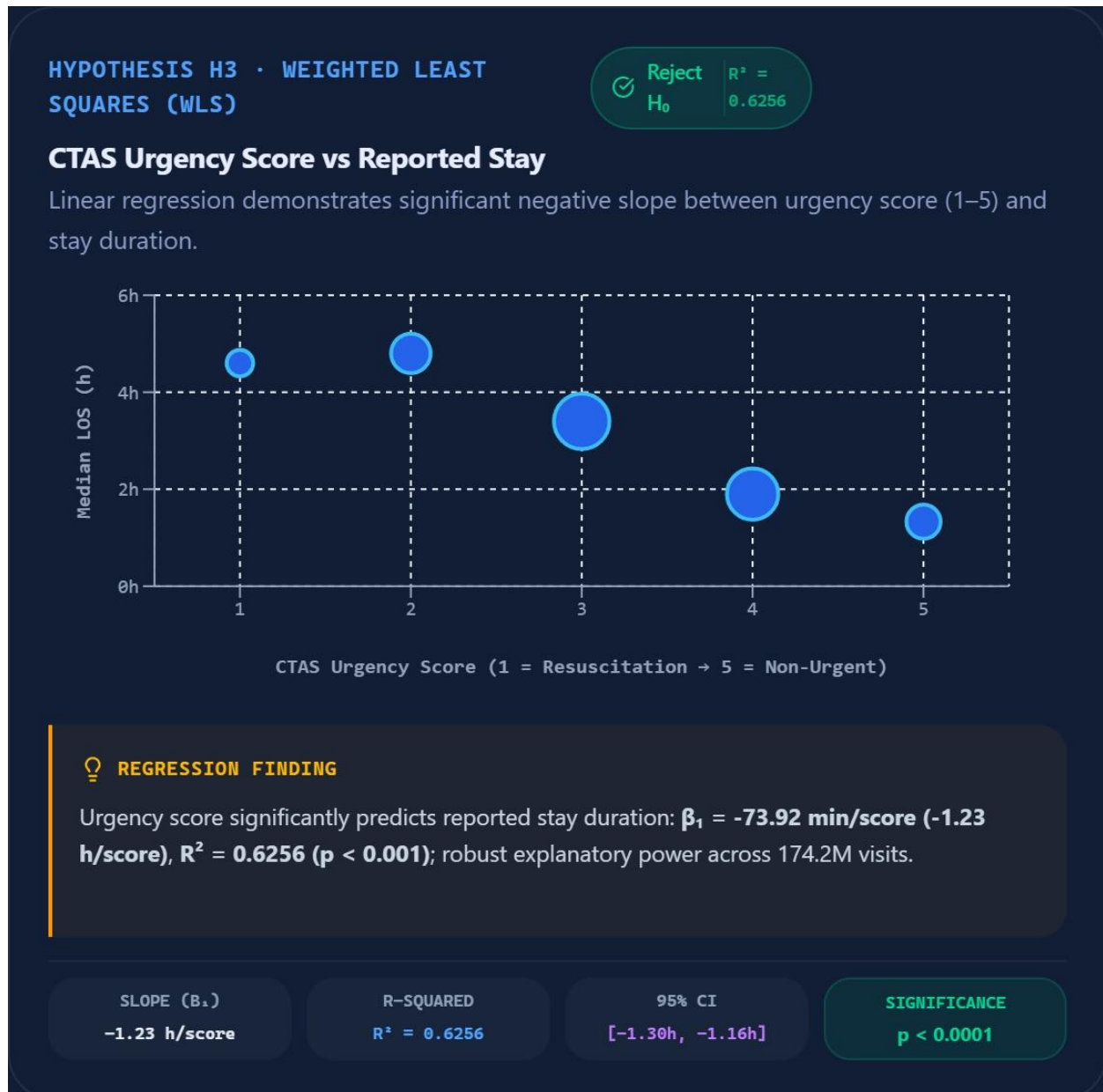
- The weighted Mann–Whitney U test produced $U = 1,124,848,860,923.5$, $p < .0001$, with a rank-biserial correlation of $r_b = .9846$.
- The difference in the two reported medians is about 3.30 hours.
- This is one of the best operational results of the project. The length of time spent in the ED for admitted visits is significantly more than the length of time spent in the ED for non-admitted visits.
- The discovery suggests that there is a strong relationship between ED throughput and downstream hospital processes. Patients are permitted to remain in the ED treatment areas after they are admitted pending bed placement in the inpatient unit.

Decision-Support Implication

Interventions that involve the entire hospital are necessary to improve the throughput of EDs, not just in the ED.

H3 — Multivariable LOS Model

Figure 15. Multivariable Model of Reported ED LOS



The H3 visualization is the weighted least-squares predictive model for reported median ED LOS for the project.

The model is based on three important analytical dimensions:

- CTAS category
- Age group
- Visit disposition

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1. There are 10 aggregate observations with a model that contains 9 encoded predictors and an adjusted $R^2 = .8837$.
2. The adjusted R^2 value is high, suggesting good explanatory ability in the overall data. Results confirm the hypothesis that the reported LOS is a multi-dimensional concept and not only based on triage acuity.
3. The model should be understood, however, as an aggregate explanatory model, rather than as a model for the individual patient. The relatively small number of aggregate observations compared with the number of encoded predictors limits the extent to which the model can be generalized to individual patients.

Decision-Support Implication

Predictive systems should be made more complex with significantly larger datasets, ideally at the patient level, prior to the use of the system for individual-level clinical prediction.

H4 — Age Group and Reported LOS

Figure 16. Age Group vs. Reported LOS



The H4 visualization examines differences in reported median LOS across four age cohorts.

Age Cohort	Reported Median LOS
Pediatric & Youth	2.02 h
Young Adults	2.47 h
Middle Adults	2.73 h
Older Adults	4.01 h

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- The weighted Kruskal–Wallis test produced $H = 126,863,835.837$, $p < .0001$, with $\epsilon^2 = .7218$. All the 6 pairwise comparisons were statistically significant after Bonferroni correction.
- The visualization depicts an increase in the reported median LOS for the age cohorts, with the Older Adults reporting the longest median LOS.
- This is the result that identifies older people as a significant population for operational planning. Extended length of stay could be an indicator of more complex assessment, diagnostic needs, functional issues, social-care needs, and/or disposition planning.
- The analysis does not conclude that the finding is a cause and effect relationship. It is important to consider age as an important planning characteristic and not as a direct cause of higher LOS.

Decision-Support Implication

Early assessment, a coordinated diagnostic process, a functional evaluation, and disposition planning are important components of specialized older adult pathways that may benefit EDs.

The patient's gender and the type of visit they had. Gender of patient and disposition of visit.

H5 — Patient Sex and Visit Disposition

Figure 17. Patient Sex vs. Visit Disposition



The H5 visualization shows the correlation between patient sex and visit disposition.

The Pearson chi square test yielded: $\chi^2(1) = 18,164.97$, $p < .0001$

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The Cramér's $V = .0102$, however, is very small for a meaningful association.

The observed proportions of admissions are about:

- Female: 9.95%
- Male: 10.56%

The gap between the two is about 0.62 percentage points.

The outcome is a good illustration of the difference between statistical and operational significance. The association is statistically significant due to the large number of cases, but it is negligible in practical terms.

Decision-Support Implication

Based on current results, patient sex should be excluded from being an important operational intervention target. Such resources should rather be channeled to factors that have significantly higher correlations with reported LOS and resource burden.

Estimated Resource Burden by CTAS

Figure 18. Estimated Resource Burden by CTAS



- The Estimated Resource Burden visualization goes beyond visit volume and LOS to combine CTAS urgency, reported LOS, and visit volume into an acuity-weighted planning measure.
- The visualization highlights Urgent as the largest contributor (about 52.3% of total derived resource burden, or about 766.7 million acuity-hours) in CTAS III.
- The biggest contribution comes from CTAS III given its high reported LOS and high encounter volume, around 72.10 million encounters.

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- This finding highlights a key difference between clinical severity and cumulative burden of operations. The highest acuity category is CTAS I, and the low amount of cases does not contribute significantly to the total burden of the system. In contrast, CTAS III has a medium to high clinical urgency and very high volume.

Decision-Support Implication

Healthcare resource planning should consider the effect of patient numbers, length and acuity of stay, not just the most clinically severe ones.

The estimated resource-burden measure serves as an indicator of planning in the project context. It should not be read as requirements for number of staff, cost or as a clinically validated burden measure.

Top Main Presenting Problems

Figure 19. Top Main Presenting Problems by ED Visit Volume



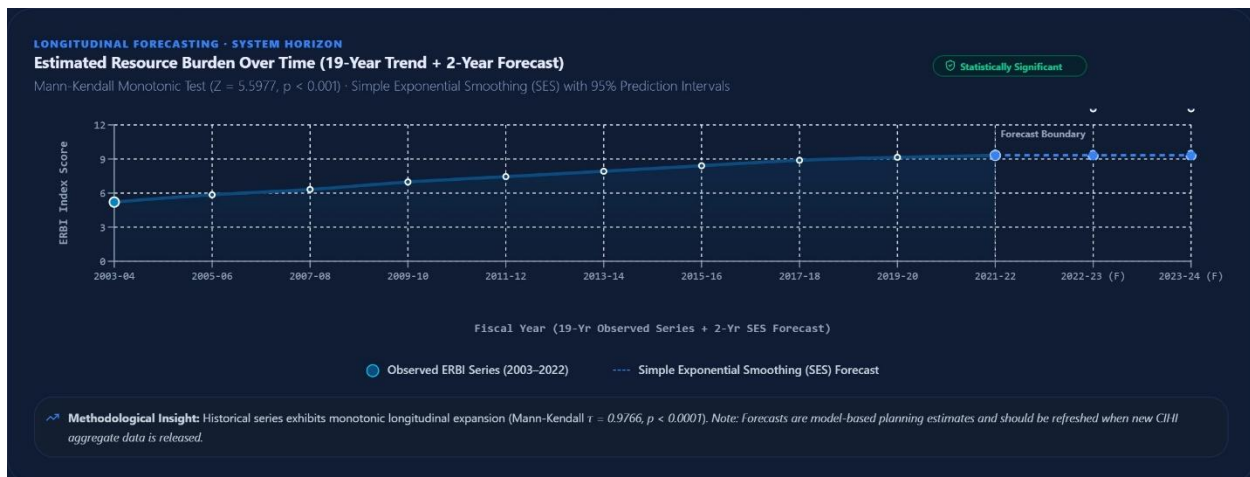
- A clinical-profiling visualization ranks selected presenting problems by the number of ED visits, and includes reported LOS context.
- Unintentional Falls (approximately 10.0 million visits) and Trauma (approximately 31.5 million visits) are the top two presenting problem categories.
- The visualization also shows that there is no correlation between high volume and long duration in terms of clinical category. Acute Myocardial Infarction is one such case, with a reported median LOS of ~4.98 hours, and a much smaller number of visits.
- The result underlines the need to take account of both the number and the length of operations when assessing operational burden.

Decision-Support Implication

Standardized/optimized care pathways may be useful for high-volume conditions, while specialized diagnostic, treatment and/or consultation resources may be warranted for lower volume conditions with longer LOS.

Estimated Resource Burden Over Time: 19-Year Trend and Two-Year Forecast

Figure 20. Estimated Resource Burden Over Time



The final visualization combines the longitudinal ERBI series with a two-year model-based forecast.

Observed ERBI increased from:

4.21 in FY 2003–2004 → 8.33 in FY 2021–2022

The Mann–Kendall analysis indicates a strong positive longitudinal association:

$\tau = .9766, p < .0001$

The forecasting model produces the following estimates:

Fiscal Year ERBI

FY 2021–2022 8.33 — Observed

FY 2022–2023 8.65 — Forecast

FY 2023–2024 8.98 — Forecast

The visualization provides a forward-looking perspective on potential resource burden. The model suggests continued short-term growth following the historical trajectory.

These values are model-based planning estimates and should not be interpreted as guaranteed future outcomes.

Decision-Support Implication

The combination of historical trend analysis and forecasting provides decision-makers with an early planning signal. The ERBI should be recalculated whenever new CIHI data become available and forecast accuracy should be evaluated against subsequently observed values.

7.4 Dashboard Audiences and Interaction Modes

The dashboard caters to three main user groups: executive decision makers, clinical operations & bed management team, data analysts.

Executive Users:

High level KPI cards, a one-click export capability for board level reporting and an ERBI capacity gauge are provided for executive users. The features enable the leadership to track the trend of the pressure in the entire ED without needing to interpret the statistical output.

Clinical Operations and Bed Management:

Clinical operations users are presented with more detailed operational information, such as an acuity-flow matrix, presenting complaints ranked by duration, and an alert on boarded admissions that last longer than eight hours. The operational monitoring tools are based on these views, which represent a translation of the statistical findings.

Data Analysts:

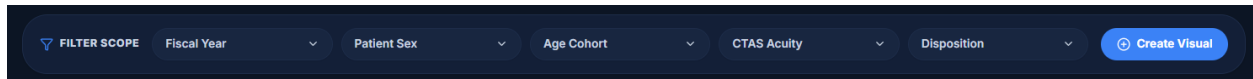
The live hypothesis-testing suite, forecasting sandbox and session-isolated upload portal allow data analysts to access the data. The forecasting sandbox allows users to play with the value of the Simple Exponential Smoothing parameter, and the isolated upload lets regional data or other data be explored without changing the baseline.

7.5 Interactive Filtering and Decision Support:

The visualization layer of the dashboard is identical to the analytical calculations presented in Chapters 4-6. It contains the nineteen-year time series of visit volumes, comparison with CTAS, comparison with admission status, WLS regression, forecasting view and ERBI trajectory.

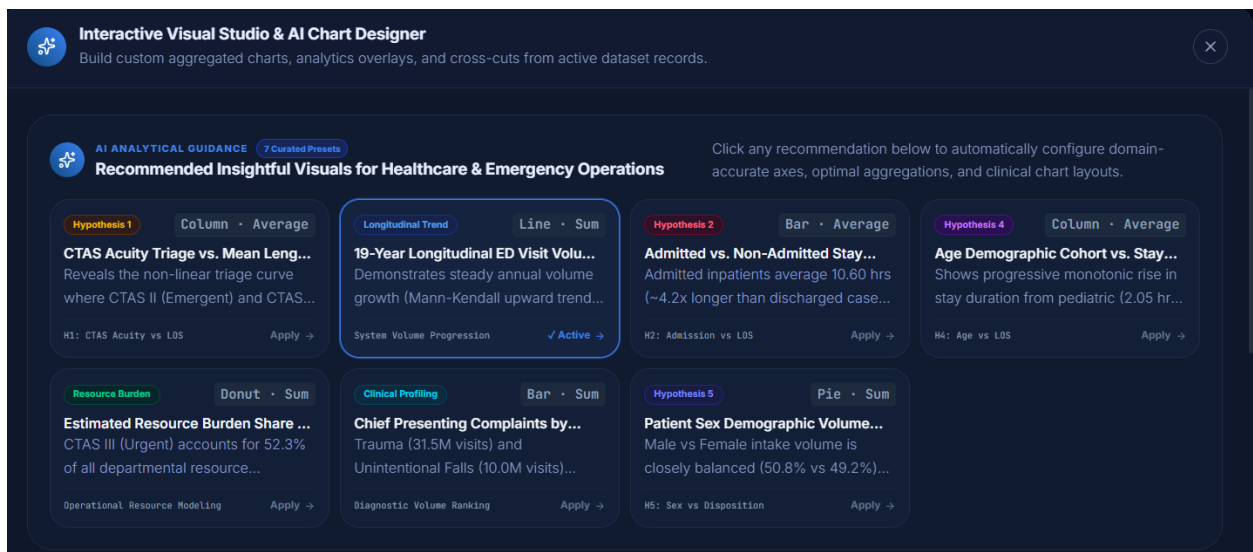
Every visualization includes a description name, labeled axis, legend (when needed) and hover level information that lets users explore individual reporting strata. This design makes the dashboard more than just a set of static charts, but an interactive analytical environment that is directly connected to the database.

Figure 18
Dashboard Filter Scope Panel

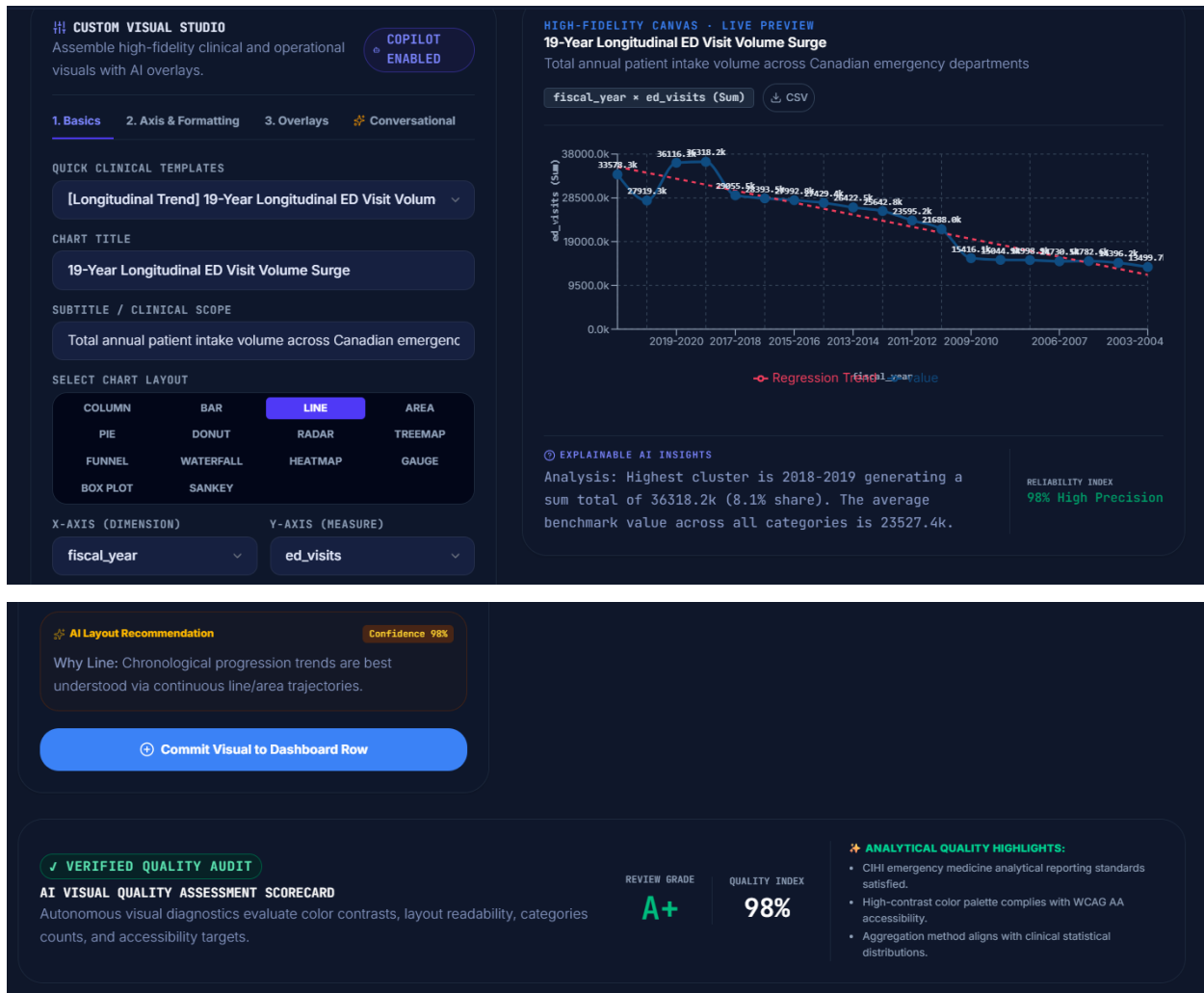


The active dataset can be limited by the filter-scope panel on five dimensions: Fiscal Year Patient Sex Age Cohort CTAS Acuity Disposition It is possible to use these filters alone or in combination prior to creating a visualization. The seven dashboard figures shown above in this chapter are therefore only a selection of the confirmed views on the system which the platform can produce. The need for visualization as a decision-support layer. The need for visualization as a decision-support layer.

7.6 Interactive Visual Studio and AI-Assisted Chart Design



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It features the Interactive Visual Studio and AI Chart Designer to assist in creating personalized analytical visuals to support the development of customizable healthcare and ED reports. By offering the users a guided environment to choose chart types, set analytical dimensions and measures, add visualization overlays and preview the visualization before adding it to the dashboard, the component will make it easier for them to reach these goals.

The visual studio makes a few suggestions depending on the variables you have for analysis. These include the visualizations of the five hypothesis tests, the volume of ED visits over time, estimated resource burden, clinical profiling and demographic analysis. This recommendation layer supports the user's decision in choosing the visualization formats, depending on the analysis question and not just manually configuring the chart.

The interface offers chart-design options such as chart title, subtitle and clinical scope, chart layout, X-axis dimension and Y-axis measure which are all configurable. The available layouts are

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column, bar, line, area, pie, donut, radar, tree map, funnel, waterfall, heatmap, gauge, box plot and Sankey visualizations. Flexibility enables the same data set to be analyzed through various ways of operation and provides a consistent analytical workflow.

In longitudinal analysis, it is recommended to perform a line-based visualization since the progression of time is best reflected by a continuous line. In the example shown, the X-axis dimension is fiscal year, and the measure is ED visits. The resulting preview shows the annual ED visit trajectory, along with a regression trend line, which allows users to evaluate the trajectory of their visits to the ED and the general direction of change.

The AI-assisted feature also delivers explainable analytical insights along with the visualization that is generated. The example shown here is a system that is used to determine the fiscal-year cluster that has the greatest fiscal-year visit volume and return a proportion of the overall data set for that cluster. It also shows the reliability of the generated visualization, to enhance transparency in the automated recommendation process.

An extra layer of validation is added by a separate AI Visual Quality Assessment Scorecard. The configuration that is shown is reviewed as A+, and it has a quality index of 98% and the analysis method has been selected appropriately, as well as accessibility requirements, number of categories, contrast and ease of reading the layout. The interface also claims to be compliant with healthcare analytics reporting standards and WCAG AA accessibility concerns.

The Interactive Visual Studio is an extension of the static dashboard reporting available with Visual Studio, and enables users to build, test and tweak analytical visualizations on the fly. AI recommendations for charts, customizable visualization settings, insights explained, and automatic quality assessment gives a flexible mechanism for how to explore healthcare operational data to make decisions.

This interactive visual studio forms an excellent way for the students to become acquainted with key insights:

- The platform offers recommendations, powered by artificial intelligence, for the choice of visualization for analytical questions.
- The user may set the title, clinical scope, layouts, dimensions and measures before publishing a visualization to the dashboard.
- A line-based trajectory is used to represent the longitudinal ED visit analysis, suitable for analyzing changes over fiscal years.

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- The system can augment visualizations with meaningful analysis and trend overlays.
- The demonstrated quality assessment has an A+ review grade and 98% quality index.
- The visual-quality assessment includes accessibility, readability, contrast and analytical-method suitability.
- The component also enhances the platform's ability to be an interactive decision-making platform instead of a reporting dashboard.

Chapter 8: Findings, Synthesis, and Critical Discussion

8.1 An Operational Triad

The 5 hypothesis tests and longitudinal analysis suggest that the operational burden of ED is not a simple phenomenon. The analysis reveals three key dimensions: clinical acuity and complexity, admission and downstream disposition, and age composition of patients.

H1 shows significant differences in reported median ED LOS among the CTAS categories. After Bonferroni adjustment, all 10 pairwise comparisons are statistically significant, with final medians of 3.20 hours for CTAS I, 3.60 hours for CTAS II, 3.70 hours for CTAS III, 2.90 hours for CTAS IV and 2.00 hours for CTAS V, and $\epsilon^2 = .7479$.

One of the best operational separations is H2. The median LOS is 6.10 hours for admitted visits, and 2.80 hours for non-admitted visits. The difference in the reported medians is 3.30 hours, and the rank-biserial correlation is .9846, which is very high, suggesting a fairly strong separation between the two groups.

Another example of the dimensionality of reported median ED LOS is given in H3. The WLS model includes the CTAS category, age group category, and disposition of visit and has an adjusted R^2 of .8837. The model was aggregate based and the results should be interpreted as aggregate based relationships and not as patient based results.

The demographic dimension is added by H4. There is an increase in reported median LOS in all three broad age cohorts, with the highest being in the Older Adults age category (4.01 hours), followed by the Pediatric & Youth age category (2.02 hours), and $\epsilon^2 = .7218$, with all six pairwise comparisons statistically significant.

In H5, it is emphasized that statistical significance and practical significance are not the same thing. The sex–disposition association is statistically significant, but it has an operationally small magnitude, as measured by Cramér's $V = .0102$.

These findings concur with a system-level approach to capacity planning that considers not only acuity and disposition, but age composition, volume, and duration when assessing the demand for ED services as reflected in a measure of LOS or the number of visits.

8.2 Comparison with Existing Literature

These findings are like those in the Canadian emergency-care literature, especially the literature reporting on the issue of emergency department overcrowding as a problem of the health system, not just the emergency department (ED). Previous studies conducted in Canada have

pointed to access block and delayed inpatient placement as key factors associated with ED crowding (Affleck et al., 2013; Li et al., 2026). This is reinforced by the present analysis that shows a large separation between admitted and non-admitted visits in the H2 analysis.

The r_b of 0.9846 for this platform suggests that the separation is not just statistically significant because of the large dataset, but also a significant separation in the reported LOS distributions. The age results are also generally consistent with previous studies that have documented the extra demands on the operational aspects of older populations attending A&E departments. The Older Adult cohort reports a median LOS of 4.01 hours, whereas Pediatric & Youth reports 2.02 hours. This difference is large, not just a function of sample size, based on the large H4 effect size in the aggregate reporting data.

The H1 results give another operating perspective. The median LOS for the platform is the longest for the CTAS III — Urgent (3.70 hours) and is close to 3.60 hours for the CTAS II — Emergent. This shows that the highest clinical acuity category doesn't always correspond to the highest aggregate throughput burden. Very high-quality patients may be stabilized and disposed immediately, while many patients who are urgent and emergent can be kept in the department to be investigated, consulted, treated and dispositioned.

The H5 result illustrates the difference between statistical and substantive significance. It is possible to obtain very small P-values for differences with little operational significance in large health care databases. The significance of the results ($p < .0001$) and the effect size (Cramér's $V = 0.0102$) indicate why effect-size measures are essential to interpret large administrative data sets.

8.3 Methodological Strengths

- The analysis was reliable and repeatable due to several methodological strengths. First, the analysis treats every visit-count observation as a valid data point, instead of treating each aggregate reporting row as an equal observation. This helps to avoid the small reporting stratum being represented by the same statistical influence as a stratum with a significantly higher number of visits to the emergency department. The resulting estimates are then closer to the actual population that the CIHI aggregate data represent.
- The use of non-parametric methods is justified in the context of emergency department LOS data, due to the lack of normality assumptions. The aggregate distributions of LOS are more appropriate compared with weighted Kruskal-Wallis and Mann-Whitney

procedures and effect-size measures are used to describe information beyond simply determining statistical significance.

- Multivariable modelling: The H3 analysis accounts for some factors other than CTAS, including age group, CTAS category, and visit disposition. The model explains 88.37% of the variance in the data. Due to the small number of observations ($N = 10$) in the model, though, and the fact that there are nine predictors encoded, the coefficients and fit statistics should be interpreted with caution and tested with a larger analytical sample before deployment for predictive purposes.
- The entire analytical process is programmed and repeatable, from the preparation of the source data to the creation of the database and computation of statistics. The project also involves around three hundred automated tests, which also constitute a layer of computational validation. Finally, the platform architecture distinguishes between a baseline research cohort and user uploaded datasets that are read-only. This enables exploratory analysis without changing the data set that the reported research results are based on, safeguarding the repeatability of the research outcomes.

8.4 Limitations

- The findings are limited by several factors. The greatest constraint is that the data in the underlying CIHI are aggregate stratum-level data, not observations of patients. Frequency weighting will improve population representation but won't transform aggregate records to individual patient records. Thus, these results should not be interpreted as representing the typical LOS of any given patient. The source tables also lack identifiers at the hospital or geographic level that can be used to model variation between hospitals or by geography.
- Thus, the analysis does not allow for conclusions to be drawn as to which facility types, provinces or individual hospitals are responsible for which patterns in the overall results. The analysis is descriptive in nature. The statistically significant relationships found in H1–H5 show association but not causation.
- For instance, the admitted/non-admitted visit difference is very large and provides for a very good correlation between disposition and LOS, but the overall analysis will not be able to independently establish each of the mechanisms for that difference. An important structural limitation of the H3 regression is also the same. The model applies the categorical

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representation of clinical complexity implemented via CTAS, age, and disposition, not a full patient-level representation.

- Thus, the CTAS representation is a planning level predictor, rather than a full clinical model. The results should only be interpreted as relationships in the overall data set, and not as a clinical prediction model. The source tables are also not populated with outcomes like mortality, readmission or ICU transfer. The project does, therefore, not measure the clinical outcomes directly, but instead assesses throughput, LOS, admission and resource burden. Another drawback is the limited period over which the observations were made (nineteen years).
- Observed changes in the number of visits over time may be due to changes in the coverage of the NACRS and participation by facilities. The preparation pipeline standardizes the source tables and filters out summary roll-up data, but it is not possible to fully differentiate between changes in national demand and changes in reporting coverage within the analysis. Lastly, the Estimated Resource Burden Index (ERBI) is not a clinical or financial metric but rather an analytical proxy used to project the burden of a project. It is appropriate to compare the relative change in acuity-weighted burden within this project, but should not be regarded as absolute staffing needs, cost or clinically validated burden.
- Regression sample-size limitation. There are 10 observations that are aggregated across the H3 WLS model, where there are 9 encoded predictors. This fairly high R^2 predictor/observation ratio reduces stability and generalizability of regression estimates. The adjusted R^2 is then a measure of descriptiveness of the model fit, not evidence of validated individual level predictive ability. The future studies should test the model with a significantly more number of aggregate observations or patient level data.
- Overall, these restrictions do not undermine the results but indicate the level of interpretation that should be made. The evidence is strongest for use as a system level planning and decision support process, and not as a single clinical prediction system.

Chapter 9: Strategic Recommendations and Implementation Roadmap

9.1 Strategic Framework

The recommendations created because of this project follow the same trajectory as the analysis results:

- Handle escalating demand for ED services.
- Monitor longitudinal LOS.
- Develop acuity-sensitive pathways.
- Address admission-related boarding.
- Strengthen multidimensional analytics.
- Develop older-adult pathways.
- Make emphasis on operationally meaningful variables.
- Prioritize resources on the basis of volume, acuity and duration.
- Identify presenting problems that occur across a large number of individuals and/or over extended periods of time and optimize them.
- Create ongoing forecasting and analytics governance.

This will help the recommendations be tied to evidence produced from the project.

9.2 Manage Increasing ED Demand

The longitudinal volume analysis shows a significant rise in the aggregate ED activity, from 4.91 million visits in FY 2003-2004 to 13.99 million visits in FY 2021-2022, with the highest number of visits being approximately 15.08 million visits in FY 2018-2019.

To make the best use of planning, therefore, it is important that the healthcare organization take into account historical utilization trends.

Recommended actions include:

- Track ED demand over the year.
- Determine if there are any obvious times when activity is higher.
- Compare the volume growth to available clinical capacity.
- Evaluate staffing and physical-space requirements.
- Use demand trends in planning for the year.
- Make forecasts about resource needs.

Reporting coverage of aggregates can vary over time, therefore results of trends should be interpreted in conjunction with data coverage information.

9.3 Monitor Longitudinal LOS

The second visualization shows that the median LOS has been generally rising throughout the study period.

Organizations should also track not just if LOS is above a target threshold, but if it is showing consistent growth.

Recommended measures include:

- Median LOS
- Higher-percentile LOS
- The time from exposure to when the child is first assessed.
- Time to treatment
- Time to disposition
- Admission-to-transfer time
- Boarding duration

A longitudinal dashboard should be employed to help detect early signs of deterioration before business or operational limits are reached.

9.4 Develop Acuity-Sensitive Care Pathways

The differences in reported median LOSs are evident in H1 with CTAS III and CTAS II reporting the longest median LOS.

Meanwhile, the resource-burden analysis shows that CTAS III is the biggest contributor to cumulative burden derived.

Acuity-sensitive pathways should thus be considered, where each pathway is divided into segments according to the clinical needs and resource use of the patient segments.

Potential interventions include:

- Rapid assessment pathways
- Clear guidelines on how to present common things.
- Uniform guidelines in common presentations.
- Facilities for suitable lower acuity patients
- Clinically appropriate nurse-initiated assessment
- Early diagnostic ordering

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- Streamlined consultation pathways
- Ongoing monitoring of the level of service (LOS) relevant to the CTAS.

The goal is not to just reduce LOS. The goal is for patients to be on the correct pathway based on their clinical requirements.

9.5 Address Admission-Related Boarding

One of the best operational results has been realized by H2.

For admitted visits, the median LOS is 6.10 hours, as compared to 2.80 hours for non-admitted visits, and the rank biserial correlation is .9846.

This means that the processes of admission warrant a lot of attention.

Recommended interventions include:

- It is not an inpatient daily discharge goal.
- It is not a daily inpatient discharge goal!
- Earlier discharge of planned discharges
- Discharge lounges
- Centralized bed management
- All beds visible in real time to patients.
- Patients can see all beds in real time.
- Admission-to-transfer monitoring
- Boarding escalation protocols
- Coordination between ED and inpatient leadership;

The primary goal should be to shorten the length of stay of patients in ED after making the clinical decision to admit.

9.6 Strengthen Multidimensional Analytics

H3 shows that reported ED LOS is correlated with several dimensions, such as CTAS, age group, and disposition. The platform's multivariable model has a strong explanatory performance within the aggregate analytical dataset, with an adjusted R^2 of .8837.

Healthcare companies must therefore not use just one variable to measure their operational performance.

An ideal analytics environment would be a fusion of:

- Acuity
- Age

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- Disposition
- Visit volume
- Presenting problem
- LOS
- Admission demand
- Inpatient capacity
- Historical trends

The current H3 model should not be used as a clinical prediction model for individual patients.

9.7 Establish an Older-Adult Emergency Management Pathway

H4 focuses on Older Adults as the group with the most time reported at the median LOS of 4.01 hours, whereas Pediatric & Youth had a median LOS of 2.02 hours.

Potential interventions include:

- Early geriatric assessment
- Geriatric liaison support
- Assessment and identification of functional needs at an early stage. Early identification of functional needs.
- Social-care assessment
- Coordinated diagnostic pathways
- Early disposition planning
- Community-care referrals
- Subacute-care coordination
- An evaluation of any avoidable delay in admission. Evaluation of any avoidable delay to admission.

9.8 Prioritize Operational Significance Over Statistical Significance

H5 is a great example of how the absence of statistical significance does not mean that an operation is not a priority.

The statistical association between sex and disposition is significant ($p < .0001$) but it has very little practical significance (Cramér's $V = .0102$).

The difference between the proportions admitted is about 62/63 of a percentage point.

This analysis should not therefore be used as a sex as an intervention variable for ED flow.

The same should be applied to other parts of the country:

- Effect size and operational relevance should be used in conjunction with statistical significance.
- This is especially crucial when processing extremely large administrative healthcare datasets.
- The goal should be to minimize unnecessary delays while assessing and disposing cases appropriately.

9.9 Allocate Resources Using Volume, Acuity, and Duration

The Estimated Resource Burden analysis shows that CTAS III represents ~52.3% of the acuity-weighted burden derived.

This discovery indicates that the focus of operational planning shouldn't be just on the number of patients.

A more extensive planning framework should be considered:

- Volume $\times$ Duration $\times$ Acuity
- This can help recognize populations which could have significant overall effects despite not being the highest acuity category.
- The outcome should be viewed as a planning proxy and is always in conjunction with recognized operational and financial measures.

9.10 Optimize High-Volume and High-Duration Presenting Problems

The presenting-problem analysis shows that the largest number of visits was for Trauma with approximately 31.5 million visits and Unintentional Falls with approximately 10.0 million visits. Meanwhile, the result in Acute Myocardial Infarction illustrates how lower volume categories can be relatively long in reported LOS (median ~4.98 hours).

This indicates two counter-productive strategies:

High-volume categories Focus on:

- Standardized pathways
- Rapid assessment
- Process optimization
- Appropriate diagnostic protocols

- Demand management

High-duration categories Focus on:

- Diagnostic coordination
- Specialist consultation
- Treatment pathway optimization
- Disposition planning
- Admission processes

The difference enables the operational teams to not see all high-volume conditions and all long-stay conditions as the same issue.

9.11 Establish Continuous Resource-Burden Forecasting

The ERBI analysis demonstrates a substantial increase from 4.21 to 8.33 over the 19-year period, with $\tau = .9766$, $p < .0001$.

The two-year forecasts of 8.65 and 8.98 provide a forward-looking planning signal.

Hospitals and health systems should therefore establish a recurring forecasting process.

The process should include:

- Update the dataset when new reporting data become available.
- Recalculate ERBI.
- Reassess the longitudinal trend.
- Refit the forecasting model.
- Compare forecasts with observed values.
- Calculate forecast error.
- Adjust planning assumptions where required.
- Communicate changes to operational leadership.

The forecasting process should support planning rather than replace professional judgment.

9.12 Strategic Implementation Roadmap

Phase	Timeline	Intervention	Primary Success Measure	Responsible Role
1	Months 1–6	ED demand and CTAS pathway optimization	CTAS-specific LOS and throughput improvement	Clinical Director, Emergency Medicine
2	Months 6–18	Inpatient discharge and bed-flow improvement	Reduced admission-to-transfer time and boarding	VP Clinical Operations / CMO
3	Months 12–18	Automated bed and boarding monitoring	Improved bed visibility and escalation	CIO / Flow Coordinator
4	Months 18–36	Older-adult emergency management pathway	Reduction in Older-Adult median LOS	Geriatric & Emergency Medicine Leadership
5	Ongoing	Resource-burden monitoring and forecasting	Forecast accuracy and sustained burden monitoring	Lead Data Scientist / Quality Committee

9.13 Measuring Implementation Success

Implementation should be evaluated using measurable outcomes.

Demand

- ED visit volume
- Seasonal/annual variation
- Forecast error

LOS

- Overall median LOS
- CTAS-specific median LOS
- Older-Adult median LOS
- Higher-percentile LOS

Admission

- Admission-to-transfer time
- Boarding duration
- Number of admitted patients occupying ED spaces

Resource Burden

- ERBI
- CTAS-specific resource burden
- Year-over-year resource-burden change

Clinical Profiling

- High-volume presenting problems
- High-duration presenting problems
- Pathway-specific LOS

Analytics

- Forecast accuracy
- Data-quality indicators
- Dashboard usage
- Frequency of analytical updates

9.14 Limitations of Implementation

The recommendations should be interpreted within the limitations of the underlying dataset.

The CIHI source is aggregate reporting data rather than individual patient-level records. Frequency weighting improves representation of reporting strata but does not transform aggregate observations into patient-level data. The dataset also does not provide sufficient hospital-level or geographic detail to determine which individual facilities or regions drive the observed patterns.

The relationships identified in H1–H5 demonstrate association rather than causation. In particular, the H3 model should not be interpreted as a validated clinical prediction model.

The ERBI is also a project-specific planning proxy and should not be interpreted as direct staffing requirements, financial cost, or clinically validated resource utilization.

These limitations reinforce the importance of combining the capstone findings with local hospital data before implementing operational changes.

9.15 Closing Note

This project demonstrates how large-scale emergency department data can be transformed into an integrated analytical and decision-support framework.

The results show that emergency department performance is influenced by multiple interconnected dimensions. ED visit volume has increased substantially over the observed period, reported LOS has demonstrated an overall upward trajectory, and significant differences exist across CTAS categories, admission status, and age groups.

Among the hypothesis tests, admission status demonstrates the strongest operational separation, while the age and CTAS analyses reveal additional dimensions of ED throughput. The sex-disposition analysis demonstrates that statistical significance does not necessarily imply operational importance.

The resource-burden analysis adds another dimension by showing that CTAS III contributes the largest share of the derived acuity-weighted burden. The presenting-problem analysis further demonstrates that high-volume and high-duration conditions require different operational responses.

Finally, the ERBI trend provides a forward-looking perspective. The increase from 4.21 to 8.33, combined with the positive longitudinal trend and model-based forecasts of 8.65 and 8.98, supports continued monitoring and forecasting as part of long-term capacity planning.

The recommended strategy is therefore based on five central priorities that is demand management, admission-flow improvement, acuity-sensitive pathways, older-adult care optimization, and continuous analytics governance.

The overall objective is not simply to reduce ED length of stay. Rather, the objective is to develop a sustainable, evidence-based approach to emergency department capacity planning in which volume, acuity, duration, disposition, demographic composition, resource burden, and future demand are considered together.

This approach transforms the project from a retrospective descriptive analysis into a practical decision-support framework for healthcare operational planning.

References

- Affleck, A., Parks, P., Drummond, A., Unger, B., & Ovens, H. (2013). Emergency department overcrowding and access block. *Canadian Journal of Emergency Medicine*, 15(6), 359–370. <https://doi.org/10.2310/8000.2013.130954>
- Alnahari, A., & A'aqoulah, A. (2024). Influence of demographic factors on prolonged length of stay in an emergency department. *PLOS ONE*, 19(3), e0298598. <https://doi.org/10.1371/journal.pone.0298598>
- Beveridge, R., Clarke, B., Janes, L., Savage, N., Thompson, J., Dodd, G., Murray, M., Jordan, S., Warren, D., & Vadeboncoeur, A. (1998). Canadian Emergency Department Triage and Acuity Scale: Implementation guidelines. *Canadian Journal of Emergency Medicine*, 1(3, Suppl.).
- Canadian Institute for Health Information. (2024). *Emergency department crowding: Beyond primary care access*. CIHI.
- Canadian Institute for Health Information. (2025). *Emergency department wait time for physician initial assessment*. CIHI.
- Canadian Institute for Health Information. (2026). *National Ambulatory Care Reporting System (NACRS): Emergency department supplementary data tables, 2003–2022* [Data set]. CIHI.
- Carter, E. J., Pouch, S. M., & Larson, E. L. (2014). The relationship between emergency department crowding and patient outcomes: A systematic review. *Journal of Nursing Scholarship*, 46(2), 106–115. <https://doi.org/10.1111/jnu.12055>
- Cevik, M., Kavaklioglu, C., Razak, F., Verma, A., & Basar, A. (2023). Assessing the impact of emergency department short stay units using length-of-stay prediction and discrete event simulation (arXiv:2308.02730). *arXiv*. <https://doi.org/10.48550/arXiv.2308.02730>
- Conover, W. J. (1999). *Practical nonparametric statistics* (3rd ed.). John Wiley & Sons.
- Field, A. (2018). *Discovering statistics using IBM SPSS statistics* (5th ed.). SAGE Publications.
- Few, S. (2012). *Show me the numbers: Designing tables and graphs to enlighten* (2nd ed.). Analytics Press.

Capstone Final Report

- Gilbert, R. O. (1987). *Statistical methods for environmental pollution monitoring*. Van Nostrand Reinhold.
- Hyndman, R. J., & Athanasopoulos, G. (2018). *Forecasting: Principles and practice* (2nd ed.). OTexts. <https://otexts.com/fpp2/>
- Kendall, M. G. (1975). *Rank correlation methods* (4th ed.). Charles Griffin.
- Li, M. K., McLeod, S. L., Affleck, A., Bhate, T., Innes, G., Parks, P., Petrie, D. A., Rowe, B. H., Snider, C., Worrall, J. C., Mackay, F., & Ovens, H. (2026). *Emergency department overcrowding* [CAEP position statement update]. Canadian Association of Emergency Physicians. <https://caep.ca/position-statements/emergency-department-overcrowding>
- Lin, M., Lucas, H. C., Jr., & Shmueli, G. (2013). Research commentary—Too big to fail: Large samples and the p-value problem. *Information Systems Research*, 24(4), 906–917. <https://doi.org/10.1287/isre.2013.0480>
- Mirzadeh, P., Kuk, J. L., Wharton, S., Reid, R. A., & Arden, C. I. (2024). Healthcare outcomes and dispositions in persons with obesity within emergency departments in Ontario, Canada: A cross-sectional analysis of the National Ambulatory Care Reporting System (NACRS), 2018–2022. *PLOS ONE*, 19(9), e0311190. <https://doi.org/10.1371/journal.pone.0311190>
- Munzner, T. (2014). *Visualization analysis and design*. CRC Press.
- OECD. (2023). *Health at a glance 2023: OECD indicators*. OECD Publishing. <https://doi.org/10.1787/7a7afb35-en>
- Pines, J. M., Pollack, C. V., Diercks, D. B., Chang, A. M., Shofer, F. S., & Hollander, J. E. (2009). The association between emergency department crowding and adverse cardiovascular outcomes in patients with chest pain. *Academic Emergency Medicine*, 16(7), 617–625. <https://doi.org/10.1111/j.1553-2712.2009.00456.x>
- Provost, F., & Fawcett, T. (2013). *Data science for business: What you need to know about data mining and data-analytic thinking*. O'Reilly Media.
- Robinson, W. S. (1950). Ecological correlations and the behavior of individuals. *American Sociological Review*, 15(3), 351–357. <https://doi.org/10.2307/2087176>
- Singer, A. J., Thode, H. C., Jr., Viccellio, P., & Pines, J. M. (2011). The association between length of emergency department boarding and mortality. *Academic Emergency Medicine*, 18(12), 1324–1329. <https://doi.org/10.1111/j.1553-2712.2011.01236.x>

Capstone Final Report

- Tomczak, M., & Tomczak, E. (2014). The need to report effect size estimates revisited: An overview of some recommended measures of effect size. *Trends in Sport Sciences*, 21(1), 19–25.

Appendix A: System Validation and Testing Evidence

A.1 Purpose of System Validation

To evaluate the reliability of the performance of the developed EDA platform in conducting the analytical, visualization and reporting functions, system validation was conducted. The validation process primarily centered on the accuracy of the analytical outputs, consistency of the results between platform modules, usability of interactive dashboards, and separating the analytical workflows from the underlying baseline dataset.

The validation approach was structured with the key analytical elements already available in the platform, such as data preparation, exploratory analysis, hypothesis testing, trend analysis, forecasting, dashboard visualization, strategic interpretation and export of reports.

A.2 Analytical Validation

The statistical presentation of the platform was checked against the analytical methods mentioned in the methodology. The validation of the main elements related to hypothesis testing are as follows: The outputs have been consistently applied in the interpretation given in Chapter 5 and 6.

Analytical Component	Statistical Method	Key Validation Output
H1 — CTAS and ED length of stay	Kruskal–Wallis test with post-hoc pairwise comparisons	Significant differences across CTAS categories; $\epsilon^2 \approx 0.748$
H2 — Admission and ED length of stay	Mann–Whitney U test with rank-biserial correlation	Strong separation between admitted and non-admitted groups; rank-biserial correlation = 0.9846
H3 — Multivariable length-of-stay model	Multiple regression	Adjusted $R^2 = 0.8837$
H4 — Age and ED length of stay	Kruskal–Wallis test with post-hoc comparisons	Significant differences across age cohorts; $\epsilon^2 \approx 0.7218$
H5 — Sex and disposition	Chi-square test with Cramér’s V	$\chi^2 = 18,164.97$, $df = 1$, $p < .0001$; Cramér’s V ≈ 0.0102
Long-term throughput trend	Kendall’s tau	$\tau = 0.9766$, $p < .0001$
Two-year throughput forecast	Holt’s Linear Exponential Smoothing	FY+1 = 8.65; FY+2 = 8.98

A.3 Dashboard and Visualization Validation

This dashboard layer was examined to make sure the analytical results would be conveyed by interactive visual elements, and not only by statistical tables.

It offers filtering and exploring capabilities by the following dimensions:

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- Fiscal Year
- Sex
- Age Cohort
- CTAS
- Disposition

The visualization layer displays statistical results using chart, summary card, trend visualization and analytical comparison. This will enable users to shift from general descriptive data to hypothesis-testing results and strategic conclusions.

A.4 Forecast Validation

The forecasting component was assessed based upon the historical emergency-resource burden index (ERBI) series of the past 19 fiscal years from FY2003-04 to FY2021-22. The historical series rose from around 4.21 to 8.33 in FY2021–22 from FY2003–04. Mann–Kendall analysis showed a significant (p value $< .0001$) strong positive trend ($\tau = 0.9766$).

The two-year forward projection was then created using Holt's Linear Exponential Smoothing. The subsequent forecast rose from 8.65 to 8.98 for FY+2, suggesting that the upward trend in resource pressure is likely to continue, assuming the historical trend remains unchanged.

A.5 Functional Validation

The platform was also tested for functionality to ensure that all the main modules worked as a coherent analytical workflow. The platform structure implemented is composed of 7 stages:

- About Project
- Prep & Quality
- Dataset Explorer
- Hypothesis Testing
- Executive Dashboard
- Strategic Insights
- Reports & Export

This framework offers a sequence of data-quality awareness, statistical analysis, and interpretation by executives, culminating in a report.

A.6 Validation Summary

All in all, the validation evidence shows that the platform merges descriptive analytics, statistical inference, regression modelling, longitudinal trend analysis, forecasting and interactive visualization into a holistic decision support environment. The results reported in the rest of the report are derived from the outputs of the analytical functions put in place in the platform and form the basis for the recommendations of the strategic plan developed in Chapter 9.

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Appendix B: Rubric Alignment Checklist

This appendix correlates the key assessment criteria of the capstone with the sections of the report in which the evidence is found. This mapping illustrates how the project meets the analytical, methodological, technical, interpretive and communication needs of the capstone.

Rubric Criterion	Evidence in This Report
Problem Analysis and Context	Chapter 1 defines the emergency-department operational problem, healthcare context, project objectives, and decision framework. Chapter 2 establishes the analytical methodology and project workflow.
Data Collection and Preparation	Chapter 3 documents the CIHI NACRS data source, dataset structure, temporal coverage, data-grain considerations, weighting rationale, cleaning procedures, feature engineering, and data governance.
Analytical Methods and Implementation	Chapters 2 and 5 explain and implement the statistical methodology, including frequency-weighted analysis, non-parametric testing, regression modelling, and the five hypothesis tests (H1–H5).
Interpretation and Insights	Chapters 5, 6, and 8 interpret statistical findings using effect sizes and practical significance rather than relying solely on p-values.
Application of the Analytics Lifecycle	Chapter 2 describes the project's seven-stage platform workflow, while Appendix C provides evidence of the technical implementation, outputs, and methodological safeguards associated with each stage.
Data Visualization and Communication	Chapter 7 documents the interactive decision-support platform, while the embedded analytical figures and dashboard screenshots demonstrate how statistical results are communicated to decision-makers.
Professional Structure and Technical Writing	The report incorporates an executive summary, numbered chapters and subsections, tables, figures, equations, methodological discussion, limitations, references, and appendices using a formal academic structure.

Appendix C — Analytics Lifecycle Evidence Log

The project was carried out as an integrated analytics workflow instead of as stand-alone analyses. The following evidence log translates the seven stages evident in the developed analytics platform into the key activities, technical outputs and methodological safeguards.

Platform Stage	Core Activity	Technical Implementation & Output	Methodological Safeguard
1. About Project	Established the healthcare problem, project objectives, analytical scope, and decision context.	Project overview, objectives, problem definition, and analytical context presented within the platform.	Clearly defined project boundaries and distinguished the analytical focus from broader hospital-system issues.
2. Prep & Quality	Prepared and assessed the historical CIHI NACRS data before analysis.	Data-cleaning and preparation workflow; standardized analytical dataset and quality checks.	Removed inappropriate aggregate roll-up records, harmonized categories, and addressed data-quality issues before statistical analysis.
3. Dataset Explorer	Enabled structured exploration of the prepared dataset across important dimensions.	Interactive dataset exploration using fiscal year, sex, age cohort, CTAS, and disposition filters.	Maintained the distinction between aggregate reporting strata and patient-level observations when interpreting results.
4. Hypothesis Testing	Tested the five predefined research hypotheses concerning length of stay, admission, demographics, and disposition.	H1–H5 statistical analysis modules and associated result visualizations.	Applied appropriate non-parametric and categorical methods to account for skewed distributions and the structure of the aggregate data.
5. Executive Dashboard	Consolidated major analytical findings into an accessible decision-support interface.	Interactive KPI cards, statistical charts, trend visualizations, and filter controls.	Presented statistical measures alongside visual context to reduce reliance on isolated p-values or single summary statistics.
6. Strategic Insights	Translated analytical findings into operationally relevant conclusions and recommendations.	Strategic findings and implementation-oriented recommendations derived from H1–H5	Used effect sizes and practical interpretation to distinguish statistically detectable relationships from

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		and longitudinal analysis.	operationally meaningful findings.
7. Reports & Export	Converted analytical outputs into formal reporting and reusable deliverables.	Report-ready analytical outputs and export functionality.	Preserved consistency between platform results and the documented capstone findings.